



NEXUS HUBBARD CAMPUS

CARRIER HOTEL

Basis of Design

December 2025

Nexus Doc No: NHG-TC-DB-00001

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1. EXECUTIVE SUMMARY

The Nexus Hubbard Carrier Hotel is a purpose-built telecommunications facility designed to serve as the campus's network gravity center, private peering hub, and Internet Exchange Point (IXP). This document establishes the design principles, criteria, and assumptions for a 487,500 SF facility with 50 MW of IT load capacity serving approximately 3,500 cabinets across a highly diverse tenant base.

The Carrier Hotel represents critical infrastructure for the Texas corridor, aggregating 11,600 fiber strands from 13+ major carriers and creating a self-reinforcing network effect where each new network participant increases value for all tenants. The facility is designed to support the broadest range of customers from Tier 1 carriers establishing points of presence to hyperscalers deploying peering routers to enterprises requiring single-cabinet network equipment deployments.

Strategic Function

Every byte of data flowing to and from the adjacent AI Hyperscale buildings and Neocloud facilities passes through this infrastructure, establishing predictable recurring revenue streams that scale with campus growth. The facility's position as both a private peering hub and Internet Exchange Point positions Hubbard as fundamental internet infrastructure comparable to established IXPs in Dallas, Houston, and other major markets.

Regulatory Status

In support of this vertically integrated network strategy, Nexus Data Centers, LLC is pursuing designation as a Competitive Local Exchange Carrier (CLEC) and Service Provider Certificate of Operating Authority (SPCOA) from the Public Utility Commission of Texas (PUCT). This regulatory status aligns the facility's carrier-neutral interconnection infrastructure with formal telecommunications authority.

Purpose of Document

This Basis of Design Report serves as the foundation for all subsequent design work, defining parameters that impact the design process including: - Facility requirements and occupancy definitions - System types and performance criteria - Applicable codes, standards, and owner requirements - Technical specifications for all building systems.

2. DESIGN ANALYSIS

2.1 Introduction

The Carrier Hotel is designed as a single-story telecommunications facility optimized for network-centric workloads with emphasis on reliability, security, and operational flexibility. The facility accommodates extreme tenant diversity with deployment flexibility ranging from single-cabinet installations to 50-cabinet peering infrastructure consuming hundreds of kilowatts.

Key Design Parameters:

Parameter	Value
Total Capacity	50 MW IT load
White Space	487,500 SF
Rack Count	~3,500 cabinets
Tenant Count	500+ organizations
Minimum Deployment	1 cabinet
Power Density	5-15 kW/rack (network-optimized)
Cooling Strategy	Raised floor air / contained cold aisle
Water Intensity	Minimal per MW
Lease Terms	Month-to-month flexibility
Primary Functions	Private peering, IXP operations, carrier interconnection
Floor Loading	350 lbs/sq ft
Fiber Aggregation	11,600 strands from 13+ carriers

Design Priorities:

1. **Network Gravity Effect:** Dense fiber and carrier presence that attracts additional networks
2. **Security in Depth:** Layered physical and cyber-physical security protecting the Meet-Me Room core
3. **Operational Flexibility:** Month-to-month terms with scalable deployments
4. **Carrier Neutrality:** Open access model supporting competitive telecommunications
5. **Reliability:** N+1 electrical redundancy with air-cooled infrastructure for maximum uptime

3. DISCIPLINE DETAILS

3.1 CIVIL

3.1.1 Overview

The civil scope for the Carrier Hotel focuses on site integration within the broader Hubbard campus while maintaining the facility's distinct operational requirements. The Carrier Hotel requires specialized considerations for diverse fiber entry points, carrier vehicle access, and utility separation from adjacent data hall infrastructure.

3.1.2 Codes and Standards

- 2024 International Fire Code (IFC)
- 2010 Americans with Disabilities Act (ADA)
- National Fire Protection Association Codes (NFPA)
- ANSI/TIA-758 Customer Owned Outside Plant Telecommunications Infrastructure Standard
- Local Land Development and Stormwater Management Codes
- Local Building Code in effect

3.1.3 Site Layout

A. Carrier Hotel Positioning - Located strategically to serve as campus telecommunications gateway - Sited to facilitate diverse fiber entry from multiple carriers - Positioned for independent utility access separate from data halls - Adjacency to campus entry for carrier technician access

B. Fiber Entry Point Diversity - Minimum two geographically diverse fiber entry locations - Underground duct bank routing from diverse cardinal directions - Vault locations positioned for carrier hand-hole access - Protected pathways maintaining >100 ft separation between diverse routes

C. Site Roads and Access - Dedicated carrier/vendor parking area separate from general campus parking (minimum 20 spaces) - 24/7 carrier technician vehicle access without requiring escort - Minimum 26' wide fire apparatus access roads with 40' inside turning radius - Loading dock access for equipment delivery with WB-67 design vehicle accommodation - Segregated visitor and carrier technician entry points

D. Parking - Carrier Hotel staff parking: 10-15 spaces - Carrier technician parking: 20-25 spaces (24/7 accessible) - Visitor parking: 10 spaces - Total: 40-50 spaces - ADA and EV charging compliance per IBC requirements

3.1.4 Utilities

A. Telecommunications Infrastructure - Diverse fiber entry from minimum two cardinal directions - Underground duct bank systems sized for 11,600+ fiber strands - Vault system supporting 13+ major carriers with expansion capacity - Separate pathways for each carrier to Meet-Me Rooms - Direct conduit routing avoiding intermediate splice points where possible

B. Water Mains - 8" domestic water service for facility and fire suppression - 12" dedicated fire main with looped configuration for redundancy - Fire hydrant spacing per 2024 IFC requirements - Isolated from data hall water systems for independent maintenance

C. Storm Drainage - Catch basin system designed for 25-year storm event (6.5", 3.5 in/hr intensity) - 0.5% minimum slope for gravity flow - RCP, HDPE, PVC, or ductile iron piping as appropriate - Independent outfall from data hall drainage systems

D. Sanitary Sewer - 8" SDR-26 PVC main at 0.4% minimum slope - 6" SDR-26 PVC service laterals to facility - Manhole spacing at 400' maximum - Grease interceptor for kitchen facilities - Oil-water separator for loading dock floor drains

E. Electrical Service - Utility-grade duct bank from campus substation - Diverse routing separate from data hall electrical feeds where practical - Designed for 75 MW capacity (50 MW IT + 25 MW infrastructure with PUE 1.5) - Phased deployment supporting 5 construction phases of 10 MW IT load each

3.2 ARCHITECTURAL

3.2.1 Preliminary Code Analysis

A. Occupancy Classification (IBC Section 302) - Group B: Business occupancy (administrative offices, network operations center) - Group F-1: Factory Industrial (Meet-Me Rooms, equipment rooms, carrier cages)

B. Occupancy Separation - No separation required between B and F-1 in fully sprinklered building per IBC Table 508.4 - MMR designated as high-security zone with enhanced physical separation

C. Occupancy Load (IBC Table 1004.1.2) - Based on "Actual Occupancy" - facility has minimal routine occupancy - Office areas: 1 person per 150 SF - Meet-Me Rooms: 1 person per 300 SF (maintenance/technician access) - Total estimated building occupancy: 50-75 persons

D. Type of Construction - Type II-B, Fully Sprinklered - Non-combustible construction - No special fire-resistive requirements beyond sprinkler protection

E. Building Limitations - Single-story building: Unlimited area per IBC Section 507.4 - Building height: 32'-0" to underside of roof structure - Surrounded by yards minimum 60' wide on all sides - Total building area: 487,500 SF

F. Fire Resistive Building Elements (IBC Table 601) - Primary structural frame: 0 hours - Exterior/interior walls: 0 hours - Floor/roof construction: 0 hours - **Exception:** 2-hour fire barrier

for main electrical room (if required by NEC) - **Exception:** MMR core enhanced security partitions (slab-to-slab, not fire-rated requirement)

G. Fire Protection Systems - Automatic fire sprinkler system (NFPA 13): 100% coverage - Fire detection system (NFPA 72): Full coverage - Preaction systems in Meet-Me Rooms and critical equipment areas

H. Means of Egress - Maximum travel distance (Group F-1, sprinklered): 250' - Maximum common path of egress: 100' - Minimum corridor width: 44" (accessible route) - Exit door width: 36" minimum clear

3.2.2 Space Planning

A. Facility Organization

The Carrier Hotel is organized into distinct functional zones optimized for security, operational efficiency, and carrier access:

Space Type	Area (SF)	Function
Meet-Me Room A	12,000	Primary interconnection facility, carrier cages, cross-connect fields
Meet-Me Room B	12,000	Secondary/redundant interconnection facility
IXP Equipment Room	2,000	Internet Exchange Point switching fabric and servers
Carrier Cages (Multiple)	15,000	Individual carrier equipment installations
Cross-Connect Fields	8,000	High-density fiber termination and patching
Network Operations Center	2,500	24/7 staffed monitoring and operations
Carrier Tech Support Area	1,500	Workbenches, staging, temporary equipment storage
Security Operations Center	800	Integrated with NOC, dedicated security monitoring
Administrative Offices	3,200	Facility management, carrier account management
Conference Rooms	1,200	Client meetings, carrier coordination
Loading Dock	4,000	Equipment receiving, four dock positions
Equipment Staging	3,000	Pre-deployment equipment storage and testing
Electrical Rooms (Main)	4,500	Primary electrical distribution
Electrical Rooms (Secondary)	2,000	Distributed electrical closets
Mechanical Rooms	5,000	HVAC, cooling equipment

Space Type	Area (SF)	Function
Battery Rooms (UPS)	3,000	Distributed battery backup systems
Fiber Entry Rooms (2)	1,000	Diverse fiber entry points
Security Vestibules	800	Mantrap entries, access control
Restrooms/Break Areas	2,000	Staff facilities
Corridors/Circulation	16,200	Building circulation
Total	487,500	

B. Meet-Me Room (MMR) Design Principles

The Meet-Me Rooms represent the facility's core value proposition and require specialized design:

1. **Location:** Positioned in building structural core, away from exterior walls
2. **Redundancy:** Two independent MMRs with diverse fiber routing
3. **Carrier Caging:** High-density steel mesh cages (sub-0.5" openings) for individual carrier equipment
4. **Cross-Connect Density:** Dedicated fields supporting 10,000+ fiber terminations per MMR
5. **Power Distribution:** 2N power feeds to all carrier cages
6. **Cooling:** Dedicated precision cooling maintaining 72°F ±2°F, <50% RH
7. **Access Control:** Tier 4 security with biometric authentication and escort requirements
8. **Fire Suppression:** Preaction systems with aspirating smoke detection

C. Carrier Tech Work Area

Dedicated space allowing carrier technicians to: - Stage equipment prior to installation - Perform testing and troubleshooting - Access tools and test equipment - Coordinate with facility staff - Workbenches with power and network connectivity

3.2.3 Exterior Enclosure

A. Wall Construction - Insulated metal panels (IMP) by PEMB vendor - R-value: R-19 minimum (climate dependent) - Finish: Kynar 500 coating in neutral color palette - Anti-climb surface texture on lower 12' where accessible

B. Windows and Glazing - Administrative Areas: Thermally broken aluminum framing, 1" insulated low-E glazing - **Secure Areas (MMR, NOC, SOC):** No exterior windows; interior borrowed light only - **Carrier Access Areas:** Limited vision glazing for wayfinding; spandrel glass at equipment areas

C. Entry Systems - Main Public Entry: Aluminum storefront with security vestibule (mantrap) - **Carrier Technician Entry:** Separate 24/7 access point with card reader and camera coverage - **Loading Dock:** Four dock positions with levelers, seals, and bumpers - **Emergency Exits:** Panic hardware with delayed egress where permitted by code

D. Roof System - Standing seam metal roof (PEMB standard) - Minimum R-30 insulation - ¼":12" minimum slope - FM Global Class 1A-90 wind rating - Gutters and downspouts: Pre-finished metal - No roof-mounted equipment except for minimal HVAC serving office areas

E. Site Perimeter Security - K-rated anti-ram bollards at vehicle approaches - Setback distance: 50' minimum from building face to vehicle barriers - Hostile vehicle mitigation integrated into landscape design

3.2.4 Interior Construction

A. Meet-Me Room (MMR) - Floor: Raised access floor system, 24" height, 1,250 lb concentrated load rating - **Walls:** Slab-to-slab construction preventing "over-the-top" intrusion - Concrete masonry unit (CMU) or steel stud with layer of 5/8" Type X GWB each side - Partitions extend full height to structure above - **Ceiling:** Exposed structure with white painted finish, minimum 12' clear height - **Doors:** Hollow metal, 3'x7', card reader + biometric access, auto-closer, monitored contacts - **Lighting:** High-bay LED, 50 fc maintained, separately switched zones - **Finishes:** Sealed concrete or ESD-rated floor tile; painted CMU or sealed GWB walls

B. Carrier Cages within MMR - Framing: Welded steel mesh, 4-gauge wire minimum, <0.5" openings (anti-reach-through) - **Height:** Floor to deck above (full height) - **Doors:** Steel mesh with 3-point locking, card reader controlled - **Sizes:** Modular cage sizes: 100 SF, 200 SF, 400 SF, 800 SF, custom available - **Power:** Dedicated metered circuits per cage with owner-furnished panel - **Cooling:** Shared precision cooling infrastructure; tenant equipment heat load <8 kW/rack average

C. Cross-Connect Fields - Layout: High-density fiber termination racks in hot aisle/cold aisle configuration - **Pathway:** Overhead ladder rack with vertical cable managers at termination points - **Density:** Design for 96-fiber termination panels, 48 panels per bay, 20+ bays per field - **Lighting:** Task lighting at fiber termination areas, 75 fc maintained - **Access:** Controlled access requiring escort for non-facility personnel

D. Network Operations Center (NOC) - Floor: Raised access floor, 12" height, 1,000 lb rating, ESD-rated tile finish - **Walls:** Full-height GWB partitions, painted finish, acoustic treatment - **Ceiling:** 2x4 acoustical tile in suspended grid, 9'-0" AFF - **Lighting:** LED troffers, dimmable, 30 fc ambient with task lighting at workstations - **Workstations:** Modular furniture supporting dual/triple monitors, sit-stand capability - **Video Wall:** 3x3 or 4x4 display array for network monitoring and SCADA visualization - **Acoustics:** NRC 0.85 ceiling tile, sound masking system, acoustic wall treatment

E. Security Operations Center (SOC) - Adjacency: Direct visual and physical connection to NOC - **Monitoring Positions:** Minimum 4 operator positions - **Display Systems:** Multiple monitor arrays plus shared video wall - **Security Integration:** Access control head-end, video management servers, intrusion detection - **Secure Storage:** Locked evidence cabinets, secure document destruction

F. Administrative Offices - Layout: Private offices (120 SF each) + open office area (1,500 SF) - **Ceiling:** 2x4 acoustical tile, 9'-0" AFF - **Lighting:** LED troffers, dimmable, with daylight harvesting at perimeter - **Finishes:** Carpet tile, painted GWB, acoustical ceiling tile - **Windows:** Interior borrowed light to corridors; exterior windows where not facing secure areas

G. Conference Rooms - Sizes: (1) large (400 SF, 12-person) + (2) small (150 SF, 4-person) - **AV Systems:** Display screens, video conferencing capability, wireless presentation - **Acoustics:** Enhanced acoustic separation from adjacent spaces (STC 50) - **Glass:** Interior glass walls with privacy film or blinds

H. Corridors - Width: 6'-0" minimum (double-loaded), 8'-0" at main circulation spine - **Ceiling Height:** 9'-0" AFF at suspended ceiling (office areas); 12'-0"+ clear at equipment areas - **Finishes:** VCT flooring, painted GWB walls, acoustical ceiling tile - **Lighting:** LED linear fixtures, 20 fc maintained, emergency egress lighting

I. Loading Dock - Size: 4,000 SF, four dock positions - **Doors:** Overhead coiling doors, 10'W x 12'H - **Equipment:** Dock levelers (12,000 lb capacity), dock seals, dock bumpers - **Floor:** Sealed concrete, sloped to drain - **Ceiling:** Exposed structure, minimum 16' clear height - **Lighting:** High-bay LED, 50 fc maintained

J. Equipment Staging Area - Size: 3,000 SF adjacent to loading dock - **Function:** Pre-deployment testing and equipment preparation - **Floor:** Sealed concrete with integrated ground grid - **Power:** 208V/120V convenience outlets, dedicated circuits for test equipment - **Network:** Fiber and copper connectivity to test VLANs

3.2.5 Loading Dock

A. Configuration - (4) Recessed dock positions at 10' centers - Dock levelers: 12,000 lb capacity, hydraulic operation - Dock seals: Fabric-reinforced rubber, replaceable - Dock bumpers: Laminated rubber, 12"W x 10"H x 4.5"D

B. Overhead Doors - (2) High-speed fabric doors at dock approach (interior to exterior) - (4) Sectional overhead doors at dock positions (standard operation)

C. Vehicle Access - Design vehicle: WB-67 interstate tractor-trailer - Approach grade: <5% maximum - Turning radius analysis confirming unrestricted maneuvering

D. Environmental Control - Loading dock conditioned to 78°F ±2°F heating, 72°F ±2°F cooling - Vestibule configuration minimizing infiltration to adjacent conditioned spaces - Exhaust system for truck exhaust fumes

3.2.6 Fire Department Access

A. Fire Apparatus Roads - 26' minimum width, all-weather driving surface - 25' minimum inside turning radius (40' preferred for Hubbard campus standard) - 13.5' minimum vertical clearance - <10% maximum gradient (AHJ approval for steeper grades) - Loop road configuration around building perimeter

B. Building Access - Fire department access road within 50' of all exterior doors - All portions of first-story exterior wall within 250' of fire apparatus access road - Minimum (3) entry points for fire department personnel - Knox Box at main entry with building keys and access cards

C. Fire Hydrants - Spacing per 2024 IFC (typically 300-400' on centers) - Located along fire apparatus access roads - Minimum 1,500 GPM residual flow at 20 PSI per hydrant

D. Standpipe Connections - Fire Department Connections (FDC) at main entry - Clearly visible and accessible from fire apparatus road - Separate FDC for standpipe (if required) and sprinkler systems

3.2.7 Storm Shelter (Refuge Room)

Status: To be determined based on local tornado risk assessment and jurisdiction requirements. If required:

A. Design Criteria - FEMA P-361 compliant construction (not full ICC-500 storm shelter) - Precast concrete or CMU walls, concrete roof structure - FEMA-approved door assembly with fortified hardware - Capacity: Full building occupancy (75 persons) for 2-hour duration - Area: ~750 SF (10 SF/person)

B. Location - Interior room, no exterior exposure - Clear egress path from all building areas - Adjacent to main circulation corridor - Not in mechanical or electrical room

C. Equipment - Emergency lighting: Battery backup (4 hours) - Emergency communication: Two-way radio or dedicated phone line - First aid kit, emergency supplies - No furniture or non-emergency equipment stored in space

3.2.8 Signage

A. Exterior Signage - Monument sign at campus entry with Carrier Hotel identification - Building identification signage at main entry - Address numbers visible from street (12" minimum height) - Carrier technician entrance clearly marked (24/7 access) - Loading dock entrance signage - Wayfinding signs directing to main entry, carrier entrance, loading dock

B. Interior Signage - Room identification signs at all doors (ADA compliant tactile and Braille) - Wayfinding signage at main intersections and decision points - Secure area warnings ("Authorized Personnel Only," "Escort Required") - Meet-Me Room identification (MMR-A, MMR-B) - Carrier cage identification placards - Electrical room hazard signs (arc flash, shock hazard, PPE requirements) - Emergency egress path marking (photoluminescent where required)

C. Telecommunications Signage - Fiber pathway identification labels - Cable tray identification at all junctions - Equipment room labels per TIA-606-C standard - Cross-connect field rack labeling system - Carrier circuit identification at demarc points

3.3 STRUCTURAL

3.3.1 Overview

The Carrier Hotel structural design provides a robust, single-story facility optimized for network equipment loads, extensive cable tray systems, and long-term reliability. The design emphasizes flexible column-free spaces in the Meet-Me Rooms while accommodating moderate floor loading for telecommunications equipment.

3.3.2 Applicable Codes and Standards

- International Building Code (IBC), 2024
- ASCE 7-22: Minimum Design Loads and Associated Criteria for Buildings and Other Structures
- ACI 318-19: Building Code Requirements for Structural Concrete
- AISC 360-22: Specification for Structural Steel Buildings
- AISC 341-22: Seismic Provisions for Structural Steel Buildings
- TMS 402/602-22: Building Code Requirements and Specification for Masonry Structures

3.3.3 Structural Risk Category

Risk Category II per ASCE 7-22 Table 1.5-1 (standard occupancy)

3.3.4 Structural System

A. Framing Type - Pre-Engineered Metal Building (PEMB) or Conventional Steel Frame - Selection based on local material availability and cost optimization

B. Lateral System - Transverse Direction: Rigid steel frames with moment connections at eaves - **Longitudinal Direction:** Steel braced frames in selected bays - **Foundation:** Pinned base connections at columns

C. Column Spacing - Typical Bay: 30' x 40' (30' transverse, 40' longitudinal) - **Meet-Me Room Areas:** 40' x 50' for maximum clear span and equipment flexibility - **Service Areas:** 25' x 30' (standard grid)

D. Floor System - Raised Access Floor: 24" height in MMR areas, 12" in NOC/SOC - **Structural Slab:** 6" thick concrete slab-on-grade with wire mesh reinforcement - **Ground Preparation:** Vapor barrier, compacted aggregate base per geotechnical recommendations

E. Roof System - Roof Deck: Standing seam metal roof on steel purlins - **Roof Slope:** ¼":12" minimum - **Insulation:** Batt insulation between purlins, R-30 minimum - **Roof Framing:** Open web steel joists or PEMB rafters spanning to main frames

3.3.5 Loads

A. Dead Loads - Roof system (metal deck, insulation, roofing): 5 PSF - Ceiling and MEP (office areas): 4 PSF - Cable tray systems (equipment areas): 8 PSF - Raised floor system: 6 PSF

B. Live Loads

Area	Live Load
Meet-Me Rooms (raised floor)	150 PSF
Equipment rooms	125 PSF
Offices and conference rooms	50 PSF
Corridors and lobbies	100 PSF
Loading dock	250 PSF
Roof	20 PSF (minimum)
Roof equipment platforms	150 PSF

C. Floor Loading - Concentrated Loads - Raised floor system (access floor panels): 1,250 lbs concentrated load on 2.5" diameter area - Equipment racks: 3,000 lbs per rack (assuming 300 lbs per vertical foot for 42U racks) - Battery cabinets: 5,000 lbs concentrated

D. Snow Loads - Ground snow load: **Site-specific** (55 PSF used for Hubbard campus example) - Flat roof snow load: Per ASCE 7-22 calculations - Snow drift surcharges at roof projections and mechanical equipment - Minimum roof design load: 20 PSF (unbalanced snow or rain-on-snow)

E. Wind Loads - Basic design wind speed (ultimate): **115 mph** (3-second gust, site-specific) - Exposure Category: **C** - Topographic factor: 1.0 (flat terrain) - Directionality factor: Per ASCE 7-22 - Enclosed building classification

F. Seismic Loads - Seismic Design Category: **C** (site-specific, per ASCE 7-22) - Site Class: **D** (default unless geotechnical investigation specifies otherwise) - Importance Factor: 1.0 - Mapped spectral accelerations: Site-specific per ASCE 7-22 figures - Lateral force-resisting system: - Steel ordinary moment frames (R=3.5) - transverse - Steel ordinary concentrically braced frames (R=3.25) - longitudinal

3.3.6 Materials

A. Structural Steel

Member Type	Specification
Wide flange shapes	ASTM A992, Fy = 50 ksi
Hollow structural sections (HSS)	ASTM A500 Grade B/C, Fy = 46 ksi (rect.), 42 ksi (round)
Pipe sections	ASTM A53 Grade B, Fy = 35 ksi

Member Type	Specification
Angles and channels	ASTM A36, Fy = 36 ksi
Plates (connections)	ASTM A572 Grade 50, Fy = 50 ksi
High-strength bolts	ASTM A325 ($\leq 7/8"$), ASTM A490 ($\geq 1"$)

B. Concrete

Application	Compressive Strength (f'c)
Slab-on-grade	4,000 PSI
Foundations (footings, grade beams)	4,500 PSI
Equipment pads	4,500 PSI
- All concrete: Normal weight (145 PCF) - Reinforcement: ASTM A615 Grade 60 deformed bars	

C. Masonry (if used for interior secure partitions or exterior architectural features) - CMU: ASTM C90, normal weight, f'm = 2,000 PSI - Mortar: ASTM C270, Type S - Grout: ASTM C476, 2,500 PSI minimum

3.3.7 Foundations

A. Geotechnical Design Parameters (Representative - site-specific investigation required) - Allowable soil bearing pressure: 3,000 PSF - Soil density: 120 PCF - Coefficient of friction (sliding): 0.35 - Passive earth pressure: 250 PCF - Active earth pressure: 40 PCF - Frost depth: 24-36" (climate dependent)

B. Foundation Type - Spread footings at columns - Continuous strip footings at perimeter walls (where applicable) - Thickened slab edges at overhead door openings - Equipment pads for mechanical units and UPS systems

C. Underfloor Preparation - Vapor barrier: 15-mil polyethylene - Granular base: 6" minimum, compacted to 95% standard Proctor - Reinforcement: 6x6-W2.9xW2.9 welded wire mesh - Joints: Tooled control joints at 15' maximum spacing each direction

3.3.8 Deflection Criteria

Structural Element	Limit
Roof beams (live load)	L/240
Roof beams (total load)	L/180
Floor beams (live load)	L/360
Raised floor pedestals	L/500
Cantilever elements	L/180

3.3.9 Special Considerations

A. Cable Tray Support - Overhead cable tray systems: Trapeze hangers at 5' maximum spacing - Design for 50 PLF minimum distributed load - Seismic bracing per ASCE 7-22 for non-structural components ($I_p = 1.5$)

B. Equipment Attachment - Telecommunications racks: Seismically braced per ASCE 7-22 - Unistrut support systems embedded in slab or attached to structural steel - Equipment anchors: Expansion anchors or cast-in-place inserts

C. Raised Floor System - Structural support independent of building structure - Lateral bracing at perimeter and at 20' intervals - Stringers and pedestals per manufacturer's specifications for 150 PSF uniform load - 1,250 lbs concentrated load rating on 2.5" diameter area

3.4 MECHANICAL

3.4.1 Overview

The mechanical systems for the Carrier Hotel are designed to support network-centric equipment with lower heat densities than traditional data centers (5-15 kW/rack average) while maintaining precise environmental conditions critical for telecommunications reliability. The design emphasizes air-cooling strategies appropriate for the equipment profile while providing redundancy and operational flexibility.

Key Design Principles: - Air-cooled infrastructure appropriate for network equipment heat loads - Precision environmental control in Meet-Me Rooms ($72^{\circ}\text{F} \pm 2^{\circ}\text{F}$, $<50\% \text{ RH}$) - N+1 redundancy for all critical cooling systems - Energy efficiency through economization and variable capacity control - Minimal water consumption compared to high-density data centers - Independent systems for different tenant spaces allowing isolated maintenance

3.4.2 Codes and Standards

- 2024 International Mechanical Code (IMC)
- 2024 International Energy Conservation Code (IECC)
- ASHRAE Standard 62.1: Ventilation for Acceptable Indoor Air Quality
- ASHRAE Standard 90.1: Energy Standard for Buildings Except Low-Rise Residential Buildings
- ASHRAE Thermal Guidelines for Data Processing Environments (4th Edition)
- ASHRAE TC 9.9: Data Center Cooling Best Practices
- NFPA 70: National Electrical Code (for mechanical equipment electrical requirements)

3.4.3 Mechanical Systems Design Conditions

A. Outdoor Design Conditions (Representative - Abilene, TX climate) - **Elevation:** 1,790 ft - **Cooling Design (0.4%):** 101.8°F DB / 70.8°F WB - **Heating Design (99.6%):** 20.1°F DB - **Evaporative Cooling (0.4%):** 76.2°F WB

B. Indoor Design Conditions

Space Type	Cooling	Heating	Relative Humidity
Meet-Me Rooms (MMR-A, MMR-B)	72°F ±2°F	68°F ±2°F	40-50%
IXP Equipment Room	72°F ±2°F	68°F ±2°F	40-50%
Carrier Cages (occupied)	75°F ±3°F	68°F ±2°F	30-50%
Network Operations Center	72°F ±2°F	70°F ±2°F	40-55%
Security Operations Center	72°F ±2°F	70°F ±2°F	40-55%
Offices and Conference Rooms	75°F ±2°F	70°F ±2°F	40-55%
Electrical/Battery Rooms	77°F ±3°F	65°F ±3°F	<60%
Loading Dock	78°F ±5°F	55°F ±5°F	Not controlled
Corridors	75°F ±3°F	70°F ±3°F	Not controlled

C. Ventilation Rates (ASHRAE 62.1)

Space Type	Outdoor Air Requirement
Meet-Me Rooms	0.06 CFM/SF
Equipment Rooms	0.06 CFM/SF
Network Operations Center	5 CFM/person + 0.06 CFM/SF
Offices	5 CFM/person + 0.06 CFM/SF
Conference Rooms	5 CFM/person + 0.06 CFM/SF
Loading Dock	0.12 CFM/SF
Battery Rooms (sealed batteries)	Per manufacturer/NFPA requirements
Restrooms	50 CFM/water closet + 25 CFM/urinal

D. Space Pressurization - Meet-Me Rooms: +0.02" WC relative to corridors - Electrical/Battery Rooms: Neutral or slight negative to adjacent spaces - Building: +0.03" WC relative to outdoors

3.4.4 Mechanical System Description

A. Meet-Me Room (MMR) Cooling

System Type: Dedicated precision cooling with N+1 redundancy

Equipment: - **Computer Room Air Handlers (CRAH):** In-row or perimeter units - Quantity: 4 units per MMR per phase (3 operating + 1 standby = N+1); Total facility: 20 MMRs across 5 phases - Capacity per unit: 100 tons (1,200 MBH sensible) - Chilled water cooling coils (no DX compressors in critical space) - Variable speed EC fans with VFD control - Supply air temperature: 65-68°F - Configuration: Front return air, top discharge or in-row discharge to cold aisle

- **Chilled Water System:**

- Air-cooled chillers located on grade outside building
- Quantity: 3 chillers per system (2 operating + 1 standby for N+1)
- Capacity per chiller: 200 tons (per phase)
- Total system capacity per phase: 400 tons operating (600 tons installed); Full buildout: 2,000 tons operating (3,000 tons installed)
- Leaving chilled water temperature: 45°F supply / 55°F return
- Variable primary flow pumping with VFDs
- Integral water-side economizer for free cooling
- Magnetic bearing compressors (oil-free, high efficiency)

Air Distribution: - Raised access floor plenum as supply air plenum - Floor grilles in cold aisles provide conditioned air to equipment intakes - Equipment exhausts to general room volume (no hot aisle containment required at these densities) - CRAH units draw return air from room and discharge to underfloor plenum

Controls: - Supply air temperature reset based on room conditions - Chiller staging and capacity control based on system load - Economizer enable when outdoor conditions permit (typically <55°F outdoor temperature) - Humidity control via CRAH reheat or supplemental humidification

B. IXP Equipment Room Cooling

Similar to MMR cooling but scaled appropriately: - **CRAH Units:** 2 units (1 operating + 1 standby) - **Capacity:** 50 tons per unit - **Chilled Water:** Shared system with MMRs or dedicated loop depending on physical separation - **Air Distribution:** Raised floor or overhead duct supply

C. Carrier Cage Cooling

Approach: Individual carrier cages share the common MMR cooling infrastructure - No dedicated cooling equipment per cage - Heat load diversity across 500+ small tenants across all phases provides natural load smoothing - Design assumes average 8 kW/rack with peak spikes to 12 kW - CRAH capacity designed for aggregate load plus 30% diversity factor

Tenant Requirements: - Maximum heat load per cabinet: 15 kW (enforced via circuit breaker sizing) - Tenant equipment air intake from raised floor cold aisle - Tenant equipment exhaust to room return air volume - No portable cooling units permitted (maintain centralized air management)

D. Network Operations Center (NOC) / Security Operations Center (SOC) Cooling

System Type: Dedicated precision cooling with redundancy

Equipment: - CRAH Units (CRAC-style for smaller load): - Quantity: 2 units (1 operating + 1 standby) - Capacity: 20 tons per unit - Chilled water coils connected to main chilled water system - Ducted supply and return air distribution - Precision temperature and humidity control for 24/7 operations environment

Air Distribution: - Overhead ducted supply to workstation areas - Ceiling return air plenum (above suspended ceiling) - Supply diffusers: Adjustable pattern diffusers, 50 FPM maximum throw velocity - Return grilles: Above suspended ceiling at 150 SF per grille

E. Office and Conference Room HVAC

System Type: Variable Air Volume (VAV) rooftop units

Equipment: - Rooftop Units (RTU): - Quantity: 2 units (each serving approximately 5,000 SF office area) - Capacity: 15 tons cooling per unit - DX cooling with variable capacity scroll compressors - Gas-fired heating section (or electric heat in mild climates) - Variable speed supply fan - Integrated air-side economizer (100% outdoor air capable) - MERV 13 filtration minimum

Air Distribution: - VAV terminal boxes with electric reheat coils - Private offices: Individual VAV box per office or small groups of offices - Open office areas: VAV boxes serving zones not exceeding 1,500 SF - Conference rooms: Dedicated VAV box per room with CO₂-based demand control ventilation - Supply air distribution: Overhead ductwork to ceiling diffusers - Return air: Ducted return from occupied spaces

Zoning Strategy: - Perimeter zones: 15' depth from exterior wall - Interior zones: Core areas away from exterior influence - Conference rooms: Separate zones with occupancy-based control

F. Electrical and Battery Room Ventilation

System Type: Dedicated exhaust with outdoor air makeup

Equipment: - Exhaust Fans: - Quantity per electrical room: 1 fan (redundant fan for critical rooms) - Capacity: 6 air changes per hour minimum - Centrifugal roof exhaust fans with backdraft dampers - Temperature-based control: Activate at 80°F, modulate to maintain <85°F

- **Makeup Air:**
 - Powered or passive makeup air louvers
 - Outdoor air introduced at low level for natural convection cooling
 - Heated in winter to prevent freeze conditions (electric duct heater at intake)

Battery Room Considerations: - VRLA or sealed lithium batteries: Minimal ventilation required - Exhaust activated by temperature or hydrogen detection (if required by code/manufacturer) - Comply with NFPA 855 for energy storage systems if applicable

G. Loading Dock HVAC

System Type: Destratification and spot heating

Equipment: - Heating: Gas-fired unit heaters or radiant tube heaters - Capacity: Maintain 55°F minimum during winter design conditions - Intermittent operation (activated when dock doors

closed and area occupied) - **Cooling:** Destratification fans only (no mechanical cooling) - High-volume, low-speed (HVLS) ceiling fans for air circulation

Dock Seals and Door Management: - Dock seals at all dock positions minimize infiltration - High-speed fabric doors at interior/exterior transitions - Minimize door open time through operational procedures

H. Humidification System

Application: Meet-Me Rooms, IXP Equipment Room, NOC/SOC

Equipment: - **Electric Steam Humidifiers:** - Clean steam generation (no mineral deposition) - Modulating capacity control based on space humidity sensors - Multiple dispersion manifolds throughout served spaces - Auto-drain and deliming cycles - Capacity: Sized for winter design condition infiltration load plus equipment desiccation - Example MMR load: ~50 lbs/hr per MMR

Distribution: - Steam dispersion tubes mounted above raised floor or in overhead ceiling space - Multiple injection points to prevent local over-humidification - Control strategy: Maintain 40-50% RH setpoint with $\pm 5\%$ deadband

I. Economizer Operation

Free Cooling Strategy: - **Chilled Water System Economizer:** - Integrated water-side economizer on air-cooled chillers - Plate-and-frame heat exchanger bypasses chiller compressor - Operates when outdoor temperature $< 55^{\circ}\text{F}$ and cooling load present - Can provide 100% free cooling when outdoor temp $< 40^{\circ}\text{F}$

- **Office RTU Air-Side Economizer:**
 - Integrated economizer dampers on each RTU
 - Differential dry-bulb or enthalpy-based control
 - 100% outdoor air capable up to cooling design conditions
 - Minimum damper position for ventilation air during non-economizer mode

3.4.5 Equipment Specifications

A. Computer Room Air Handlers (CRAH) - Meet-Me Rooms - Type: Floor-mounted or in-row, chilled water cooling - **Cooling Capacity:** 100 tons sensible (nominal) - **Airflow:** 33,000 CFM (300 CFM/ton ratio for sensible-only cooling) - **Coil:** 6 rows chilled water, 10 FPI max, copper tubes, aluminum fins - **Fans:** EC plug fans or forward-curved centrifugal with VFD, redundant fan array - **Controls:** BACnet integration, supply air temperature sensor, return air temperature/humidity sensor - **Construction:** Insulated cabinet, front return, top or front discharge - **Filtration:** MERV 11 minimum (no internal duct lining)

B. Air-Cooled Chillers - Type: Outdoor air-cooled, magnetic bearing centrifugal compressor - **Capacity:** 200 tons nominal per chiller - **Compressor:** Magnetic bearing (oil-free), variable speed - **Condenser:** Microchannel aluminum coil with variable speed fans - **Economizer:** Integrated water-side economizer with plate-and-frame heat exchanger - **Refrigerant:** Low-GWP refrigerant (R-513A or similar) - **Controls:** Onboard DDC controller with BACnet/IP integration - **Efficiency:** Integrated Part Load Value (IPLV) > 18.0 EER - **Sound:** Sound attenuated to meet site requirements (< 60 dBA at

50 feet) - **Electrical:** 480V/3-phase - **Pumps:** Integral variable speed primary pumps within chiller package

C. Chilled Water Pumps (if not integral to chiller) - **Type:** Base-mounted vertical inline centrifugal - **Arrangement:** Variable primary flow, each chiller has dedicated pump - **Flow:** 400 GPM per 200-ton chiller (2 GPM/ton) - **Head:** 60 feet minimum (to overcome coil, piping, and control valve pressure drop) - **Motor:** Premium efficiency, VFD controlled - **Construction:** Cast iron body, bronze impeller, mechanical seal - **Accessories:** Suction strainer, isolation valves, pressure gauges

D. Rooftop Units (Office HVAC) - **Type:** Packaged DX rooftop air handling unit, variable air volume - **Capacity:** 15 tons cooling nominal - **Heating:** 200 MBH gas input (or electric resistance 50 kW) - **Supply Fan:** Plenum fan with VFD, backward-inclined, airfoil blades - **Economizer:** Integrated outdoor air economizer, differential dry-bulb control - **Filtration:** MERV 13 pleated filters - **Curb:** Factory-supplied insulated roof curb with spring isolators - **Controls:** Microprocessor-based controller with BACnet integration - **Efficiency:** Minimum 11.0 EER cooling, 80% AFUE heating

3.4.6 Piping System Design Criteria

A. Chilled Water System

System Configuration: - Variable primary flow (no secondary loop) - Reverse return piping for hydraulic balance (where practical) - Two-pipe system (supply and return)

Design Parameters: - Supply temperature: 45°F - Return temperature: 55°F - ΔT : 10°F - Flow rate: 2.0 GPM/ton (typical) - Piping velocity: 4 ft/sec maximum (2-3 ft/sec target for reduced erosion) - Pressure drop: 4 ft per 100 ft maximum

Materials: - **Above-grade piping (>2"):** Schedule 40 black steel pipe, welded fittings, 150 PSI rating - **Above-grade piping (≤ 2 "):** Type L copper tube, brazed fittings, 150 PSI rating - **Below-grade piping:** HDPE SDR-11 with fusion welded joints (if required for underground routing)

Insulation: - **Indoor:** 1.5" thick glass fiber with all-service vapor barrier jacket (ASJ) - **Outdoor:** 1.5" thick glass fiber with aluminum/polymer weather jacket, seams on bottom - **Valves/Fittings:** Preformed insulation covers

System Components: - **Expansion Tank:** Diaphragm type, sized for system volume and temperature range - **Air Separator:** High-efficiency tangential air separator at high point of system - **Strainers:** Y-strainers at all pump suctions and coil inlets - **Control Valves:** 2-way modulating control valves at each coil with electronic actuators - **Balancing Valves:** Manual or automatic flow balancing valves at each coil branch - **Isolation Valves:** Ball valves <3", butterfly valves ≥ 3 " - **Pressure Gauges:** At pump suction/discharge, coil inlet/outlet - **Temperature Sensors:** Supply/return mains, coil inlet/outlet

Freeze Protection: - Heat tracing on outdoor piping, valves, and drain lines - Glycol not anticipated (system operates year-round, heat trace for idle equipment)

B. Condensate Drainage

- **Materials:** PVC Schedule 40 DWV pipe and fittings
- **Slope:** 1/4" per foot minimum toward floor drains or indirect waste connections

- **Traps:** P-trap at each CRAH unit, sized for maximum negative pressure (2x coil pressure drop)
- **Termination:** Indirect waste to floor drains with air gap

3.4.7 Ductwork Design Criteria

A. Duct Construction

- **Material:** Galvanized steel sheet metal
- **Gauge:** Per SMACNA standards for pressure class
 - Low-pressure systems (<2" WC): SMACNA rectangular duct
 - Medium-pressure systems (2-6" WC): SMACNA rectangular duct, reinforced
- **Sealing:** SMACNA Seal Class A (all duct systems)
- **Joints:**
 - **Longitudinal:** Standing seam, Pittsburgh lock, or continuous weld
 - **Transverse:** Flanged and gasketed connections (duct >12" in any dimension)

B. Duct Sizing

System Type	Velocity	Pressure Drop
Supply air (RTU to VAV boxes)	1,500 FPM max	0.10" WC / 100 ft
Supply air (VAV boxes to diffusers)	1,200 FPM max	0.08" WC / 100 ft
Return air ducts	1,200 FPM max	0.08" WC / 100 ft
Outdoor air / economizer ducts	1,000 FPM max	0.08" WC / 100 ft
Exhaust ducts	1,500 FPM max	0.10" WC / 100 ft

C. Ductwork Insulation

Application	Insulation Type	Thickness
Supply air ducts (conditioned spaces)	Fiberglass duct wrap (FSK facing)	1.5"
Supply air ducts (unconditioned spaces)	Fiberglass duct wrap (ASJ facing)	2.0"
Outdoor air ducts	Fiberglass duct wrap (ASJ facing)	2.0"
Return air ducts (below 65°F)	Fiberglass duct wrap	1.0"
Exhaust ducts	None (unless condensation occurs)	-

Internal duct liner is not permitted (except for dedicated sound attenuation where required and approved)

D. Duct Accessories

- **Volume Dampers:** Provided at all branch takeoffs for air balance
- **Fire Dampers:** UL-555 rated, installed at all rated wall/floor penetrations

- **Smoke Dampers:** UL-555S rated, installed per mechanical code requirements
- **Combination Fire/Smoke Dampers:** Where both functions required
- **Access Doors:** Provided at all dampers, coils, and sensors for maintenance access
- **Flexible Connections:** 4" minimum length at all fan connections to isolate vibration

3.4.8 Vibration Isolation and Seismic Restraint

A. Vibration Isolation

All rotating equipment (fans, pumps, chillers) shall be provided with: - **Spring isolators:** Deflection based on equipment RPM - <600 RPM: 1.5" deflection - 600-1200 RPM: 2" deflection - >1200 RPM: 2.5" deflection - **Neoprene pads:** For equipment <1/2 HP or seismically restrained

Flexible Connections: - **Duct:** Neoprene-coated fabric, 4" minimum length - **Piping:** Braided stainless steel flexible connectors, 12" minimum length

B. Seismic Restraint

All mechanical equipment and distribution systems shall be seismically restrained per: - ASCE 7-22 Chapter 13: Seismic Design Requirements for Nonstructural Components - **Component Importance Factor (Ip):** 1.5 for equipment serving Meet-Me Rooms - **Standard Importance Factor (Ip):** 1.0 for other equipment

Equipment Restraint: - Four-way seismic restraint for floor-mounted equipment - Lateral and vertical restraint for suspended equipment (cable tray, piping, ducts) - Seismic snubbers at changes in direction or branch connections

3.4.9 Testing, Balancing, and Commissioning

A. Air Systems Testing and Balancing

- Performed by independent NEBB or AABC certified test and balance (TAB) agency
- All systems tested and balanced to $\pm 10\%$ of design airflow
- Individual diffusers and grilles balanced to $\pm 10\%$ of design
- Report submitted for owner review and approval

B. Hydronic Systems Testing and Balancing

- All pumps tested and balanced to design flow $\pm 5\%$
- All coils and terminal units tested to $\pm 10\%$ of design flow
- System flushed and chemically cleaned prior to balance
- Water treatment program established prior to system operation

C. Functional Performance Testing (FPT)

- Performed on all control sequences
- Verify temperature control, humidity control, economizer operation
- Test redundancy and failover sequences

- Emergency shutdown and restart procedures verified

D. Integrated Systems Test

- Simulate full building load with portable load banks if needed during commissioning
- Verify coordination between mechanical, electrical, and controls systems
- Document all sequence testing and deficiency resolution

3.5 CONTROLS

3.5.1 Building Management System (BMS) Overview

The Carrier Hotel shall employ a comprehensive Building Management System (BMS) integrated with a centralized campus SCADA platform. The BMS will provide direct digital control (DDC) of all HVAC systems, integration with fire alarm and security systems, and data reporting for operational optimization.

System Architecture: - Distributed DDC controllers at mechanical equipment - BACnet/IP native communication protocol - Ethernet backbone network (fiber where distance >300 ft) - Integration with campus SCADA via BACnet gateway - Local operator workstations in Network Operations Center (NOC) - Web-based remote access for authorized personnel

3.5.2 BMS Integration Requirements

A. Connected Systems - Precision cooling equipment (CRAHs, chillers, pumps) - Office HVAC (rooftop units, VAV boxes) - Ventilation systems (exhaust fans, makeup air units) - Humidification systems - Fire alarm system (monitoring only - no control) - Electrical power monitoring system (via Modbus or BACnet gateway) - Security system (door status, access events) - Lighting control system

B. Data Collection and Trending - All temperature, humidity, pressure sensors trended at 5-minute intervals - Equipment status (on/off, alarms) trended on change-of-state - Energy meters (electric, chilled water) trended at 15-minute intervals - Historical data retention: 2 years minimum

C. Alarm Management - Tiered alarm priority (Critical / Major / Minor) - Alarm notification via: - BMS operator workstation (audible and visual) - Email/SMS to on-call personnel - Integration to campus SCADA alarm system - Alarm acknowledgment and resolution tracking

Alarm Classification for Carrier Hotel: - **Critical:** Loss of cooling to Meet-Me Room, chiller failure with no backup available, fire alarm activation - **Major:** Single CRAH failure (N+1 redundancy maintained), high/low temperature alarm, humidity out of range - **Minor:** Filter dirty alarm, low flow condition, scheduled maintenance reminder

3.5.3 DDC Control Panels

A. Panel Construction - NEMA 1 rated enclosure for indoor installation - Lockable hinged door with viewing window - DIN-rail mounted components - Powered from UPS-backed circuits (one panel per UPS circuit to avoid single-point failure)

B. Controller Specifications - BACnet/IP native controller - Minimum 32-bit processor - Minimum 64 MB RAM, 128 MB flash storage - Universal inputs (analog/digital configurable) - Analog outputs: 0-10 VDC or 4-20 mA - Digital outputs: Dry contact relays, 2A minimum rating - Onboard web server for local access and diagnostics

C. Power - 120 VAC input from UPS-backed panel - Internal 24 VAC transformer for field devices (actuators, sensors) - Separate 24 VDC power supply for controller in critical applications

3.5.4 Temperature and Humidity Control Sequences

A. Meet-Me Room Precision Cooling

Primary Control Loop: 1. CRAH units modulate supply air temperature to maintain space temperature setpoint (72°F ±1°F) 2. Space temperature sensor (average of 3 sensors distributed throughout MMR) 3. Supply air temperature reset: - Base setpoint: 65°F - Reset higher if space temperature drops below 70°F - Reset lower if space temperature exceeds 74°F (alarm condition) 4. Chilled water valve modulates to maintain supply air temperature setpoint

Staging and Capacity Control: 1. Lead-lag-standby rotation of CRAH units 2. Lead unit operates continuously 3. Lag unit stages on if space temperature >73°F for 5 minutes 4. Standby unit remains off unless lead or lag unit fails (N+1 redundancy) 5. Units rotate weekly for even wear

Fan Control: - Variable speed control based on space temperature - Minimum fan speed: 40% (maintain minimum airflow for equipment ventilation) - Maximum fan speed: 100%

Humidity Control: - Humidifier activates if space RH <40% - Humidifier modulates to maintain 45% RH setpoint - High humidity alarm if >55% RH for 15 minutes - Low humidity alarm if <35% RH for 15 minutes - Dehumidification via chilled water coil with reheat (if required - rare in this climate)

B. Chiller Plant Control

Chiller Staging: 1. Lead-lag-standby rotation (3 chillers per system, 2 operating + 1 standby) 2. Lead chiller starts when chilled water supply temperature >50°F and any load present 3. Lag chiller stages on if: - Lead chiller at >85% capacity for 5 minutes, OR - Chilled water supply temperature >48°F for 10 minutes 4. Chillers stage off in reverse order when load decreases 5. Chillers rotate weekly

Water-Side Economizer: - Enable when outdoor temperature <55°F - Integrated plate-and-frame heat exchanger bypasses compressor - Can provide 100% free cooling when OAT <40°F - Economizer disabled if outdoor temperature >60°F

Pump Control: - Variable primary flow: Pump speed modulates to maintain differential pressure setpoint - Differential pressure sensor at furthest coil from chiller - Setpoint: 15 PSID minimum - Minimum flow bypass valve prevents chiller low-flow condition

C. Office VAV System Control

RTU Supply Air Control: - Supply air temperature setpoint: 55°F (adjustable 50-60°F) - Modulating cooling capacity (DX compressor stages + hot gas bypass or VFD scroll compressor) - Modulating heating capacity (gas valve or SCR electric heat control)

Economizer Control: - Differential dry-bulb economizer logic - Enable when outdoor temperature < (return air temperature - 2°F) - Damper modulates to maintain supply air temperature with minimal mechanical cooling - 100% outdoor air capable when economizer active and outdoor conditions favorable - Minimum damper position: 20% (meet ventilation requirements during non-economizer mode)

VAV Terminal Box Control: - Space temperature sensor in each zone - Setpoint adjustable 72-76°F cooling, 68-72°F heating - Primary air damper modulates to maintain space temperature - Electric reheat coil stages on if space temperature < setpoint and primary damper at minimum position - Minimum airflow setpoint: 30% of maximum (ventilation requirement)

Supply Fan VFD Control: - Duct static pressure sensor 2/3 distance into duct system - Setpoint: 1.0" WC (adjustable) - VFD modulates fan speed to maintain static pressure as VAV boxes modulate

Demand Control Ventilation (Conference Rooms): - CO₂ sensor in each conference room - Increase VAV box minimum airflow if CO₂ > 1000 ppm - Return to normal minimum airflow if CO₂ < 800 ppm

3.5.5 Equipment Scheduling and Occupancy Control

A. Occupied/Unoccupied Schedules

System	Occupied Schedule	Unoccupied Operation
Meet-Me Room Precision Cooling	24/7 continuous	Same
Chiller Plant	24/7 continuous	Same
NOC/SOC Precision Cooling	24/7 continuous	Same
Office HVAC	Mon-Fri 7:00 AM - 7:00 PM	Setback to 80°F cooling / 65°F heating
Electrical Room Ventilation	On-demand (temperature-based)	Same
Loading Dock Heating	On-demand (temperature-based + door status)	Same

B. Override Controls - Manual override buttons at each VAV zone for temporary occupancy - Override duration: 2 hours - Holiday schedule programmed into BMS (no office HVAC on holidays)

3.6 PLUMBING PROCESS

3.6.1 Codes and Standards

- 2024 International Plumbing Code (IPC)

- 2024 Uniform Plumbing Code (UPC) where adopted locally
- NSF/ANSI 61: Drinking Water System Components - Health Effects
- NSF/ANSI 372: Drinking Water System Components - Lead Content
- ASSE Standards for backflow prevention and water tempering devices

3.6.2 Domestic Water

A. System Description - Domestic water service from campus water main (see Civil section) - Water meter and reduced pressure backflow preventer in below-grade vault or above-grade enclosure at property line - Service line size: 3" minimum (fire sprinkler demand separate) - Domestic water serves: plumbing fixtures, break room appliances, mechanical equipment makeup water, hose bibs - No booster pump anticipated (verify with water flow test; provide if residual pressure <50 PSI at highest fixture)

B. Domestic Hot Water - Electric storage water heater for office/break room sinks and restrooms - Capacity: 80 gallon, 240V, 12 kW heating elements - Temperature setting: 120°F at heater outlet - Thermostatic mixing valve at fixture groups to limit outlet temperature to 110°F (burn protection) - Hot water circulation: Not required (compact layout, minimal dead legs)

C. Water Heater Location - Mechanical room or dedicated water heater closet - Floor drain provided with indirect waste connection - Temperature and pressure relief valve piped to floor drain - Electrical disconnect and dedicated circuit

D. Distribution Piping

Materials: - **Above-grade cold water:** Type L copper tube with wrought copper fittings, brazed joints - **Above-grade hot water:** Type L copper tube with wrought copper fittings, brazed joints - **Below-grade (if required):** Type K copper tube or HDPE SDR-11 with brass fittings - **Solder:** Lead-free 95-5 tin-antimony solder

Insulation: - **Hot water piping:** 1" thick fiberglass pipe insulation with ASJ facing - **Cold water piping (indoors):** 3/4" thick closed-cell elastomeric foam (prevent condensation)

Design Criteria: - Maximum velocity: 8 FPS cold water, 4 FPS hot water - Minimum pipe size: 3/4" for branch lines, 1/2" for fixture supplies - Isolation valves at all branches and equipment connections - Air chambers or water hammer arrestors at all quick-closing valves

E. Hose Bibs - Provided at exterior building perimeter: Spacing 150' maximum - Provided at mechanical rooms and loading dock - **Exterior hose bibs:** Freeze-resistant wall hydrants with vacuum breaker and removable key - **Interior hose bibs:** Standard sill cocks with vacuum breaker

F. Equipment Connections - Makeup water connections for cooling systems (see Mechanical section) - Eye wash stations: Tempered water (tepid) via ASSE 1071 thermostatic mixing valve - Emergency shower (if required): Tempered water (tepid) via ASSE 1071 thermostatic mixing valve - Humidifier fill: Cold water with solenoid valve and strainer

3.6.3 Sanitary Waste and Vent

A. System Description - Gravity drainage to campus sanitary sewer system (see Civil section) - All plumbing fixtures and floor drains connected to sanitary waste and vent system - Building drain sized for total fixture units per IPC Table 710.1(2) - Minimum slope: 1/4" per foot (1/8" per foot acceptable for pipes >3" diameter)

B. Fixtures and Floor Drains

Fixture Type	Quantity	Location	Drainage Connection
Water closets	4	Restrooms	3" waste, 2" vent
Urinals	2	Restrooms	2" waste, 1.5" vent
Lavatories	4	Restrooms	1.5" waste, 1.25" vent
Break room sink	1	Break room	1.5" waste, 1.25" vent
Mop sink	1	Janitor closet	3" waste, 2" vent
Floor drains (4" dia.)	8	Mechanical rooms, electrical rooms, loading dock	4" waste, 2" vent
Floor drains (2" dia.)	4	Restrooms, break room, entry vestibules	2" waste, 1.5" vent
Roof drains	12	Roof (interior drains at low points)	4" storm waste (separate system)
Indirect waste receptors	4	Mechanical rooms (CRAH condensate, humidifier drains)	2" receptor, 1.5" waste

C. Piping Materials

Sanitary Waste and Vent (Above Grade): - **Piping:** Service weight cast iron soil pipe, hubless, ASTM A888 - **Couplings:** Heavy-duty stainless steel no-hub couplings with neoprene gaskets - **Horizontal piping:** Minimum 1/4" per foot slope

Sanitary Building Drain (Below Grade): - **Piping:** Extra-heavy cast iron soil pipe with hub and spigot joints, ASTM A74, OR - **Piping:** PVC Schedule 40 DWV pipe, ASTM D2665 (if permitted by code) - **Cleanouts:** Cast iron with gasketed access covers at changes in direction and 50' maximum spacing

Storm Drainage (Separate System): - **Piping:** Service weight cast iron or PVC Schedule 40 DWV - **Roof drains:** Cast iron body with integral gravel guard and leaf strainer - **Overflow drains:** Provided at all roof drains, set 2" above primary drain inlet

D. Venting Strategy - Individual vents for each fixture where practical - Common vents for back-to-back fixtures (e.g., restrooms) - Vent stack through roof with increaser at roof penetration - Vent terminals minimum 12" above roof surface and 10' from air intakes

E. Indirect Waste Connections - CRAH unit condensate drains terminate to indirect waste receptors (floor sinks) - Air gap minimum 2x pipe diameter, 1" minimum - Humidifier drains piped to indirect waste receptors with air gap - Tempered drain water from equipment (cooling systems): Indirect waste with air gap

3.6.4 Plumbing Fixtures

A. Fixture Specifications

Fixture	Specification
Water Closets	Wall-hung, vitreous china, elongated bowl, 1.1 GPF (high-efficiency), sensor-activated flush valve
Urinals	Wall-hung, vitreous china, 0.125 GPF (high-efficiency pint flush), sensor-activated flush valve
Lavatories	Under-counter mount, vitreous china, sensor-activated faucet with pre-mixed supply (TMV), 0.35 GPM aerator
Break Room Sink	Stainless steel drop-in, dual bowl, gooseneck faucet with spray, manual control, 1.5 GPM aerator
Mop Sink	Terrazzo or fiberglass, floor-mounted, service faucet with vacuum breaker
Drinking Fountain	Wall-mounted, sensor-activated, chilled unit, ADA compliant, lead-free components

B. Accessories - Paper towel dispensers (touchless) - Soap dispensers (touchless) - Toilet tissue dispensers - Waste receptacles - Grab bars at accessible toilet compartments (ADA)

C. ADA Compliance - Accessible toilet compartments per 2010 ADA Standards - Mounting heights for lavatories, accessories, and controls per ADA - Clear floor space at all fixtures

3.7 FIRE PROTECTION

3.7.1 Overview

The Carrier Hotel fire protection system is designed to provide comprehensive automatic suppression throughout the facility while addressing the unique challenges of telecommunications infrastructure protection. The system emphasizes protection of high-value network equipment in Meet-Me Rooms while maintaining carrier-neutral access and minimizing water exposure to active telecommunications systems.

3.7.2 Construction Codes

- 2024 International Building Code (IBC)
- 2024 International Fire Code (IFC)
- Local Fire Marshal amendments

3.7.3 Reference Standards

- NFPA 13: Standard for the Installation of Sprinkler Systems (2022)
- NFPA 20: Standard for the Installation of Stationary Pumps for Fire Protection (2022)
- NFPA 24: Standard for the Installation of Private Fire Service Mains and Their Appurtenances (2020)
- NFPA 25: Standard for Inspection, Testing, and Maintenance of Water-Based Fire Protection Systems (2023)
- NFPA 72: National Fire Alarm and Signaling Code (2022)

3.7.4 Water Supply

A. Site Fire Water Supply (See Civil section 3.1.4 and campus plan) - Dual utility connections from campus fire water loop - Diverse routing maintaining >100 ft separation - Each connection capable of supplying full fire sprinkler demand - Minimum residual pressure at base of riser: 15 PSI at design flow

B. Fire Water Demand - Design area: 5,000 SF (Ordinary Hazard Group 2 per NFPA 13) - Design density: 0.20 GPM/SF - Sprinkler demand: 1,000 GPM - Hose stream allowance: 500 GPM (2 x 250 GPM for interior hose streams) - **Total demand:** 1,500 GPM at 15 PSI residual - Duration: 90 minutes per NFPA 13

C. Fire Pumps (if required based on available pressure) - Electric motor-driven horizontal split-case fire pump - Capacity: 1,500 GPM at 100 PSI total head (if boosting required) - Controller: Listed fire pump controller with automatic start on pressure drop - Power: Dedicated feeder from normal power source with disconnecting means ahead of main service disconnect - Jockey pump: Maintain system pressure, prevent fire pump nuisance starts - Installation: Per NFPA 20, supervised, within 2-hour fire-rated fire pump room

3.7.5 Fire Service and Risers

A. Building Service - Underground ductile iron fire service main enters building at two locations (diverse routing) - Pipe size: 6" minimum per riser - Post-indicator valve (PIV) with tamper switch at each service entrance - Backflow prevention: Double-check valve assembly (if required by water purveyor)

B. Fire Riser Rooms - (2) fire riser rooms serving separate building zones for redundancy - Room size: 100 SF minimum, dedicated use, no other services - Room separation: Located at opposite ends of building - Access: Directly from building corridor, not through secure spaces

C. Fire Department Connections (FDC) - (2) FDCs, one at each fire riser room location - Location: Within 100 ft of fire apparatus access road, visible and accessible - Configuration: Siamese connection with 2.5" NH threads (or as required by AHJ) - Check valve: To prevent backflow into fire department connection - Signage: "Fire Department Connection - Automatic Sprinkler System"

3.7.6 Sprinkler System Types and Distribution

The Carrier Hotel employs a combination of sprinkler system types based on space criticality and water exposure risk:

A. Wet Pipe Sprinkler System (Office Areas, Corridors, Restrooms, Break Room)

System Description: - Conventional wet pipe system with water-filled piping at all times - Simplest, most reliable system appropriate for non-critical areas - Automatic activation upon sprinkler head operation

Design Density: - **Office/Corridors (Light Hazard):** 0.10 GPM/SF over 1,500 SF - **Loading Dock (Ordinary Hazard Group 2):** 0.20 GPM/SF over 2,500 SF

Sprinkler Heads: - Quick-response (QR) pendent or recessed in office areas (ceiling-mounted) - Standard response upright in exposed service areas - Temperature rating: 155°F (ordinary)

B. Preaction Sprinkler System (Meet-Me Rooms A & B, IXP Equipment Room, Carrier Tech Work Area)

System Description: - Double interlock preaction system providing maximum protection against accidental water discharge - Dry pipe system (compressed air or nitrogen) upstream of preaction valve - Requires two independent events to release water: (1) Fire detection system activation, AND (2) Sprinkler head operation

Advantages for Telecommunications: - Prevents water discharge from accidental head damage or pipe failure - Time delay allows for detection system investigation before water release - Maintains dry pipes in spaces with high-value equipment

Design Density: - **Ordinary Hazard Group 2:** 0.20 GPM/SF over 2,500 SF (assumes cable/equipment fuel load)

System Zones: - MMR-A: Dedicated preaction zone - MMR-B: Dedicated preaction zone - IXP Equipment Room: Dedicated preaction zone - Carrier Tech Work Area: Shared zone with office wet pipe or dedicated preaction (TBD based on equipment value)

Preaction Valve Assembly: - Electric/pneumatic deluge valve with trim - Supervised control valve upstream and downstream (with tamper switches) - Pressure switches: High air/low air supervision on dry pipe side; water flow switch on wet pipe side - Air compressor and/or nitrogen generator maintaining supervisory air pressure - Accelerator (if pipe volume >500 gallons) to meet actuation time requirements - Control panel: Listed releasing control panel integrated with fire alarm system

Sprinkler Heads: - Dry pendent or upright (no water in head until system trips) - Temperature rating: 155°F (ordinary) - Spacing per NFPA 13 for Ordinary Hazard

C. Dry Pipe Sprinkler System (Loading Dock, if unheated or subject to freezing)

System Description: - Dry pipe valve maintains pressurized air in piping - Water releases when air pressure drops due to sprinkler head opening - Used where piping is subject to freezing temperatures

Design Density: - **Ordinary Hazard Group 2:** 0.20 GPM/SF over 2,500 SF

Sprinkler Heads: - Upright or pendent - Temperature rating: 155°F (ordinary)

System Supervision: - Low air pressure switch - High air pressure relief - Dry pipe valve trip test connection

3.7.7 Sprinkler System Materials and Installation

A. Piping Materials

Application	Material	Schedule
Underground fire service	Ductile iron	Cement-lined, mechanical joint or push-on joint
Wet pipe system (>2")	Black steel	Schedule 40 (threaded), Schedule 10 (grooved)
Wet pipe system (≤2")	Black steel	Schedule 40, threaded
Preaction system (all sizes)	Black steel	Schedule 40 (minimize corrosion in dry system)
Dry pipe system (all sizes)	Black steel	Schedule 40

Fittings: - Threaded fittings for pipe ≤2" - Grooved mechanical fittings for pipe >2" - Grooved couplings: AWWA C606 standard

B. Sprinkler Heads

Area	Head Type	Temperature	Finish
Office areas (ceilings)	QR pendent, recessed	155°F	White
Corridors	QR pendent	155°F	White or chrome
MMR / Equipment rooms	Dry pendent	155°F	Chrome
Loading dock	Upright	155°F	Black

Coverage: - Light Hazard: 130-200 SF per head (office areas) - Ordinary Hazard: 100-130 SF per head (equipment areas, loading dock)

C. Installation Requirements

- All piping pitched to drain valves (1/4" per 10 ft minimum)
- Hangers and supports per NFPA 13 (maximum spacing 12 ft for steel pipe)
- Seismic bracing per ASCE 7-22 and NFPA 13 Chapter 9 (four-way bracing for branch lines >2")
- Inspector's test connection at most remote point of each system
- Auxiliary drain for preaction and dry pipe systems (drain to floor drain or outdoors)
- Minimum 18" clearance below sprinkler heads (36" preferred in equipment areas)

3.7.8 Fire Detection System Integration

Preaction sprinkler systems require integration with automatic fire detection system (see Fire Alarm section 3.9). Key integration points:

A. Detection Methods in Preaction Areas - Aspirating smoke detection (ASD) system in Meet-Me Rooms (primary detection) - Spot-type smoke detectors at ceiling level (backup detection) - Both detection systems monitored by fire alarm control panel (FACP)

B. Releasing Logic - **Detection Alarm 1:** Single detector activation = Alarm notification only (investigate, no water release) - **Detection Alarm 2:** Cross-zone detection (two detectors in same zone activate) = Preaction valve solenoid energized - Preaction valve trips, water fills dry piping up to sprinkler heads - Audible/visual notification: "Sprinkler System Activated" - **Water Flow:** Sprinkler head opens due to heat = Water discharges, full fire alarm evacuation initiated

C. Manual Release - Manual pull station at each preaction zone allows fire department to manually trip preaction valve - Typically located adjacent to preaction valve assembly

D. Abort Feature - Not typically provided in data center environments (once detection activates preaction, water fills pipes) - Early detection gives personnel time to respond before head activation

3.7.9 System Supervision and Monitoring

All fire protection system components shall be electrically supervised and monitored:

A. Monitored Conditions - **Water Flow Switches:** Alarm on water flow in any sprinkler system zone - **Valve Tamper Switches:** Supervisory signal if any control valve moves from fully open position - **Preaction Valve Status:** Normal/tripped position monitored - **Air Pressure Switches:** High air/low air supervisory signals (preaction and dry pipe systems) - **Fire Pump Status:** Running, power failure, controller trouble - **FDC Check Valve:** Tamper switch on check valve (ensure valve not closed)

B. Annunciation - Fire alarm control panel (FACP) in main fire riser room - Remote annunciators at building entrance and security operations center (SOC) - Audible and visual signals for alarm, supervisory, and trouble conditions

C. Notification - Local alarm notification appliances (horns/strobes) per NFPA 72 - Automatic notification to central monitoring station (if applicable) - Notification to building management system (BMS) for status monitoring

3.7.10 Standpipe System (If Required)

A. Standpipe Requirement - Required if building highest floor level >30 ft above lowest level of fire department vehicle access (IBC 905.3.1) - Carrier Hotel is single-story:
Standpipe not anticipated unless required by local AHJ

B. System Type (If Provided) - Class I manual wet standpipe system per NFPA 14 - 2.5" hose connection valves in exit stairwells or at designated locations - Combined with automatic sprinkler system (fed from same risers)

3.8 ELECTRICAL

3.8.1 Overview

The electrical distribution system for the Carrier Hotel is designed to provide robust, reliable power delivery to approximately 3,500 network equipment cabinets across a highly diverse tenant base. The design emphasizes N+1 component redundancy, modularity to accommodate wide-ranging deployment sizes (single cabinet to 50-cabinet installations), and operational flexibility supporting month-to-month lease terms.

Key System Characteristics: - **Total IT Load:** 50 MW (approximately 3,500 cabinets at 5-15 kW/rack average) - **Infrastructure Load:** ~10 MW (mechanical, house power, lighting) - **Total Building Load:** ~75 MW - **Primary Distribution Voltage:** 480Y/277V, 3-phase, 4-wire + ground - **Secondary Distribution Voltage:** 208Y/120V, 3-phase, 4-wire + ground - **Reliability Target:** N+1 redundancy for critical power infrastructure - **Tenant Circuit Configuration:** 2N power feeds to individual tenant cages/cabinets

3.8.2 Codes and Standards

- International Building Code (IBC), 2024
- National Electrical Code (NFPA 70), 2023 Edition
- IEEE Standards (Grounding, Power Quality, Protection Coordination)
- NFPA 780: Standard for the Installation of Lightning Protection Systems
- NFPA 70E: Standard for Electrical Safety in the Workplace
- NETA Standards: Acceptance Testing Specifications for Electrical Power Equipment and Systems
- UL Standards for electrical equipment

3.8.3 Service and Load Criteria

A. Utility Service - Primary service from campus 34.5 kV distribution system - Dual diverse utility feeds providing 2N service redundancy - Each service capable of supporting full building load - Service entrance: Pad-mounted switchgear or outdoor metal-clad switchgear

B. Load Summary

Load Category	Capacity
Meet-Me Rooms (MMR-A, MMR-B)	5 MW total (2.5 MW each)
IXP Equipment Room	500 kW
Carrier Cages/Cabinets (distributed)	4 MW
Network Operations Center	100 kW
Security Operations Center	50 kW
Mechanical Systems (Cooling)	1.5 MW

Load Category	Capacity
House Power (Lighting, Receptacles)	200 kW
Total IT Load	10 MW
Total Building Load	~75 MW

C. Design Voltage - Primary Distribution: 480Y/277V, 3-phase, 4-wire + ground - Secondary Distribution: 208Y/120V, 3-phase, 4-wire + ground - Motor Loads: 480V, 3-phase, 3-wire + ground - Lighting: 277V, 1-phase, 2-wire + ground (equipment areas); 120V (office areas) - Receptacles: 120V, 1-phase, 2-wire + ground

D. Power Quality Targets - Voltage regulation: $\pm 5\%$ maximum at any point - Total Harmonic Distortion (THD): $< 5\%$ voltage, $< 20\%$ current - Power factor: > 0.95 (capacitor correction where required for large motor loads)

3.8.4 Electrical Distribution System Architecture

A. Medium Voltage Distribution (34.5 kV)

Service Entrance: - (2) Primary pad-mounted switchgear (A-side and B-side) - Each switchgear fed from diverse campus 34.5 kV sources - Main incoming circuit breaker with protective relaying - (6) Feeder breakers per switchgear: - (2) Feeders to MMR-A and MMR-B transformers - (2) Feeders to IXP/Carrier Cage transformers - (1) Feeder to mechanical distribution transformer - (1) Feeder to house power transformer

Medium Voltage Transformers: - Dry-type cast coil transformers rated for 34.5 kV primary - Indoor installation within dedicated electrical rooms - Sizes: 3,000 kVA to 7,500 kVA per transformer (scaled for 50 MW phased deployment) - Impedance: $\geq 5.75\%$ to limit fault current - Secondary voltage: 480Y/277V

B. Low Voltage Distribution (480V)

Main Distribution Switchboards: - UL 891 listed switchboard, dead-front construction - Located in main electrical rooms adjacent to transformers - Main circuit breaker: 4,000A frame with electronic trip unit - Feeder breakers to sub-distribution panels throughout building - Power monitoring and metering system integrated

Sub-Distribution Configuration: - MMR-A and MMR-B: Dedicated 480V switchboards with (2) 2,000A bus sections - IXP Equipment Room: 480V distribution panel, 800A main - Carrier Cage Distribution: Multiple 480V panels distributed throughout MMR spaces - Mechanical Distribution: 480V switchboards for HVAC equipment, chillers, pumps

C. Tenant Power Distribution

Carrier Cage / Cabinet Power: - Each carrier cage provided with dedicated power panel - Panel configuration: 208Y/120V, 3-phase, 4-wire + ground - Panel rating: 400A typical (scalable based on cage size) - Circuit breakers: 2-pole and 3-pole, 20A to 100A ratings - **Dual-Feed Design (2N):** Each cabinet receives two power feeds from independent sources - A-side feed from A-side electrical

distribution - B-side feed from B-side electrical distribution - Cabinet-level automatic transfer switch (ATS) or static transfer switch (STS) for seamless failover

Power Outlets: - Cabinet PDUs: Rack-mounted PDUs with (16) to (24) C13 outlets per cabinet - Cabinet power: (2) 30A or (2) 60A circuits per cabinet (one from A-side, one from B-side) - Locking connectors (L6-30 or L21-30) at cabinet base

Metering: - Revenue-grade metering at each tenant panel - Real-time power monitoring via EPMS system - kWh accumulation for tenant billing

D. UPS Systems

Critical UPS (Meet-Me Rooms, IXP, NOC/SOC): - UPS Configuration: N+1 parallel redundant UPS modules - UPS Type: Online double-conversion, transformer-free design - Module Capacity: 500 kW per module - System Capacity: Per phase: MMR: 3 MW (6 modules per 10 MW phase); Total 5 phases: 15 MW UPS capacity; IXP: 2.5 MW (5 modules for full campus) - Battery Runtime: 10 minutes at full load (design for utility transfer) - Battery Type: Valve-regulated lead-acid (VRLA) or lithium-ion (based on space availability) - Output Voltage: 480V (feeds distribution switchboards) - Bypass: Internal static bypass plus external maintenance bypass

House UPS (Fire Alarm, Security, BMS, Telecom): - UPS Configuration: Single UPS with battery backup - Capacity: 20 kVA - Battery Runtime: 4 hours minimum - Output Voltage: 208Y/120V

E. Emergency/Standby Power Systems

F. Lighting and Receptacle Distribution

Lighting Panels: - 277V lighting circuits for high-bay LED fixtures in MMRs and equipment areas - 120V lighting circuits for office and support areas - Panel ratings: 225A typical - Circuit protection: 20A lighting circuits, 15A receptacle circuits

Receptacle Panels: - 120V duplex receptacles throughout office and support areas - GFCI protection in restrooms, break rooms, loading dock - Isolated ground (IG) receptacles at workstations in NOC/SOC

Emergency Lighting: - Egress lighting powered by central inverter system - LED exit signs with battery backup (integral or central inverter) - Emergency lights in corridors, stairwells, exit pathways

3.8.5 Electrical Equipment Specifications

A. Medium Voltage Switchgear - Type: Metal-clad, indoor or outdoor - Voltage Rating: 34.5 kV, 60 Hz - BIL: 200 kV - Interrupting Rating: 40 kA symmetrical (minimum) - Circuit Breakers: Vacuum type with stored energy mechanism - Control Power: 125 VDC from station battery system - Protective Relaying: Microprocessor-based multifunction relays - Metering: Digital meters with power quality monitoring - Communication: BACnet or Modbus integration to SCADA

B. Medium Voltage Transformers - Type: Dry-type, cast coil - Voltage: 34.5 kV delta primary / 480Y/277V secondary - Ratings: 3,000 kVA to 7,500 kVA - Impedance: 5.75% minimum - Temperature Rise: 115°C / 80°C (optional 150°C / 115°C for higher ambient) - Sound Level: NEMA ST-20 standard - Terminals: Cable lugs (primary and secondary)

C. Low Voltage Switchboards - Type: UL 891 listed, dead-front, front accessible - Voltage: 480Y/277V, 3-phase, 4-wire + ground - Bus Rating: 2,000A to 4,000A - Main Breaker: Insulated case or power circuit breaker, 100% rated - Feeder Breakers: Molded case, 80% or 100% rated, electronic trip - Short Circuit Rating: 65 kA minimum - Ground Fault Protection: Provided on main and feeders >1,000A - Metering: Multifunction digital power meters on main and feeders

D. UPS Systems - Type: Online double-conversion, IGBT-based, transformer-free - Efficiency: >96% in double-conversion mode, >99% in eco-mode - Input Voltage: 480V $\pm 10\%$ - Output Voltage: 480V $\pm 1\%$ - Frequency: 60 Hz ± 0.05 Hz - Overload Capacity: 125% for 10 minutes, 150% for 1 minute - Bypass: Automatic static bypass plus manual maintenance bypass - Battery: VRLA or lithium-ion, seismically braced, temperature-compensated charging - Communication: SNMP, Modbus TCP, BACnet (to EPMS)

E. Panelboards - Type: NEMA 1 (indoor), bolt-on circuit breakers - Bus: Copper, fully rated - Main Breaker or Main Lugs: Per application - Branch Breakers: Thermal-magnetic or electronic trip, UL 489 listed - Neutral/Ground Bars: Separate, full-size with terminal lugs - Door-in-Door: Hinged panel cover with directory

3.8.6 Grounding and Lightning Protection

A. Grounding Electrode System - Building grounding electrode system per NEC Article 250 - Components: - Concrete-encased electrode (Ufer ground) in foundation - Ground ring: #2/0 AWG bare copper, buried 30" below grade, encircling building perimeter - Ground rods: 5/8" x 10' copper-clad steel, minimum (8) rods, spacing 20' on center - Structural steel: Bonded to grounding electrode system - Water service: Bonded to grounding electrode system (if metallic)

B. Equipment Grounding - All feeders, branch circuits, and busway include full-size equipment grounding conductor (EGC) - Raceway/conduit may not serve as sole equipment grounding means - Green insulated or bare copper EGC in all circuits - Bonding jumpers at all equipment enclosures

C. Telecommunications Grounding - Telecommunications Grounding Busbar (TGB) in MMR-A and MMR-B - TGB: 1/4" x 4" copper bus, tin-plated, minimum 6 feet long - TGB bonded to building grounding electrode system via #3/0 AWG conductor - TGB bonded to nearest electrical equipment ground bus - Individual rack grounding: #6 AWG to overhead ground bus in cable tray

D. Lightning Protection System - Air terminals (lightning rods) on roof at perimeter and high points, spacing per NFPA 780 - Down conductors: #2 AWG copper, minimum (4) down conductors, bonded to ground ring - Bonding: All roof penetrations, metal bodies, equipment bonded to lightning protection system - Ground ring integrated with building grounding electrode system - UL Master Label System per UL 96A

3.8.7 Cable and Raceway Systems

A. Medium Voltage Cable - Type: Single conductor, copper, EPR insulation, 35 kV rated, Type MV-105 - Shield: Copper tape, overlapping - Jacket: PVC, suitable for wet/dry locations - Installation: In underground ductbank (concrete-encased PVC conduit) or cable tray

B. Low Voltage Power Cable (600V) - Feeders >600V: Single conductor, copper, THHN/THWN, 75°C rated - Feeders <600V: THHN/THWN in conduit or Type MC cable (where permitted) - Branch Circuits: THHN/THWN in conduit or Type MC cable - Minimum Size: #12 AWG (branch circuits), #10 AWG (feeders)

C. Raceway Systems - Conduit: Rigid metal conduit (RMC), intermediate metal conduit (IMC), or electrical metallic tubing (EMT) - Exposed dry locations: EMT - Concealed locations: EMT or MC cable - Underground or wet locations: RMC, IMC, or PVC Schedule 40/80 - Cable Tray: Ladder-type or solid-bottom, aluminum or steel, with covers where required - Meet-Me Rooms: Overhead ladder tray with fiber management - Mechanical Rooms: Open ladder tray for power cables - Busway: Plug-in busway or feeder busway for high-density distribution (where applicable)

D. Color Coding - Phase Conductors: A-phase (Black), B-phase (Red), C-phase (Blue) - Neutral: White or gray - Equipment Ground: Green or bare copper - Isolated Ground: Green with yellow stripe

3.8.8 Lighting Systems

A. Lighting Design

Space	Lighting Type	Illumination Level (fc)	Control
Meet-Me Rooms	LED high-bay, 5000K	50	Occupancy sensor + manual override
IXP Equipment Room	LED high-bay	50	Occupancy sensor
Carrier Cages	LED linear, 4000K	30	Zoned switching
Network Operations Center	LED troffer, dimmable, 4000K	50 (desk), 30 (ambient)	Daylight harvesting + dimming
Security Operations Center	LED troffer, dimmable	50 (desk), 30 (ambient)	Dimming control
Offices	LED troffer, 4000K	50	Occupancy sensor + daylight harvesting
Conference Rooms	LED troffer, dimmable	50	Occupancy sensor + dimming
Corridors	LED linear	20	Always on, with emergency lighting

Space	Lighting Type	Illumination Level (fc)	Control
Mechanical Rooms	LED vapor-tight	30	Manual switch (no auto-off per hazard)
Loading Dock	LED high-bay	30	Photocell + occupancy sensor
Exterior Building	LED wall packs	10 (entries), 5 (perimeter)	Photocell + time clock

B. Emergency and Egress Lighting - Emergency Lighting Inverter: Central 3-phase 480V inverter system - Capacity: 20 kVA - Battery: 90-minute runtime minimum - Feeds designated emergency lighting fixtures and exit signs - Exit Signs: LED, red or green legends, <5 watts per sign - Emergency Lights: LED fixture heads, wall-mounted or ceiling-mounted, in all egress paths

C. Lighting Controls - Occupancy Sensors: Ceiling-mounted passive infrared (PIR) or dual-technology - Daylight Harvesting: Photocells at perimeter offices with 0-10V dimming - Time Clock: Astronomical time clock for exterior lighting - Manual Overrides: Wall switches in all occupied spaces for override capability - Integration: Lighting control system integrated with BMS for scheduling and monitoring

3.8.9 Fire Alarm and Life Safety Integration

A. Electrical Interfaces to Fire Alarm System - Power: Dedicated 120V circuits from house UPS to each fire alarm panel - Elevator Recall: Shunt trip of elevator power upon fire alarm (if elevators present) - HVAC Shutdown: Shutdown of air handlers upon smoke detection (via BMS integration) - Door Holders: Release of magnetic door holders upon alarm - Emergency Lighting: Automatic activation via inverter system

B. Mass Notification / Emergency Communication - Speakers and strobes throughout building powered by fire alarm system - Emergency voice capability from FACP in main fire riser room

3.8.10 Testing and Commissioning

A. Acceptance Testing (per NETA standards) - Medium voltage equipment: Insulation resistance, contact resistance, protective relay testing, trip testing - Low voltage equipment: Insulation resistance, circuit breaker testing, ground fault testing - UPS systems: Load bank testing at 50%, 100%, and 125% capacity - Battery systems: Load testing, capacity verification - Grounding system: Ground resistance testing (<1 ohm required for telecommunications) - Lighting: Illumination level verification (footcandle measurements)

B. Functional Testing - UPS failover and bypass operation - Lighting controls and occupancy sensors - EPMS data acquisition and alarming

C. As-Built Documentation - Single-line diagrams showing as-built configuration - Panel schedules with final circuit loading - Cable schedules and routing drawings - Test reports from NETA and commissioning authority - Operation and maintenance manuals

3.9 FIRE ALARM

3.9.1 Overview

The Carrier Hotel fire alarm system provides comprehensive life safety protection with emphasis on early warning detection in telecommunications spaces and coordination with preaction sprinkler systems protecting high-value network equipment.

3.9.2 Construction Codes

- 2024 International Building Code (IBC)
- 2024 International Fire Code (IFC)
- Local Fire Marshal amendments

3.9.3 Reference Standards

- NFPA 72: National Fire Alarm and Signaling Code (2022)
- NFPA 70: National Electrical Code
- UL 864: Control Units for Fire Alarm Systems
- UL 268: Smoke Detectors for Fire Alarm Systems
- UL 521: Heat Detectors for Fire Alarm Systems

3.9.4 Fire Alarm System Architecture

A. System Type - Addressable, intelligent fire alarm system - Distributed network architecture with peer-to-peer communication - Primary command and control from main FACP in fire riser room - Sub-control panels in distributed electrical rooms

B. Fire Alarm Control Panels (FACP) - Main FACP: Located in main fire riser room - Capacity: 2,000+ addressable points - Integrated voice evacuation capability - Redundant Class A signaling loops - Onboard IP network interface for remote monitoring - Sub-Panels: Located in secondary fire riser room and electrical rooms - Networked to main FACP via fiber optic backbone (Class X survivability)

C. Annunciators - Main Building Entry: Graphic annunciator with floor plan display and microphone - Security Operations Center: Full-feature annunciator with alarm acknowledgment capability - Carrier Technician Entry: LED annunciator indicating alarm/supervisory/trouble

3.9.5 Initiating Devices

A. Aspirating Smoke Detection (ASD) - Meet-Me Rooms - Technology: Very Early Smoke Detection Apparatus (VESDA) or equivalent - **Coverage:** MMR-A, MMR-B, IXP Equipment Room - **Sampling Points:** Maximum 400 SF per sampling point - **Pipe Network:** Concealed above raised floor and at ceiling level - **Sensitivity:** Four-level alarm (Alert, Action, Fire 1, Fire 2) - **Integration:** Fire 2 alarm triggers preaction sprinkler release - **Monitoring:** ASD network interface to main FACP for point-level reporting

B. Spot-Type Smoke Detectors - **Coverage:** Office areas, corridors, NOC, SOC, mechanical rooms, electrical rooms - **Type:** Photoelectric smoke detectors, addressable - **Spacing:** Per NFPA 72 (typically 30' spacing in smooth ceiling areas) - **Locations:** Ceiling-mounted throughout non-ASD protected areas

C. Duct Smoke Detectors - **Coverage:** All air handling units and rooftop units >2,000 CFM - **Type:** Photoelectric sampling-type duct detectors - **Function:** Shutdown air handler upon detection, prevent smoke distribution - **Interface:** Hardwired relay to air handler controls

D. Heat Detectors - **Coverage:** Loading dock, electrical rooms, mechanical rooms (as supplemental to smoke detection) - **Type:** Fixed-temperature or rate-of-rise, addressable - **Rating:** 135°F (ordinary) or 200°F (high) based on ambient conditions

E. Manual Pull Stations - **Locations:** At all building exits, maximum 200' travel distance per NFPA 72 - **Type:** Single-action pull stations (unless otherwise required by AHJ) - **Quantity:** Minimum 12 stations throughout building

F. Waterflow and Supervisory Devices - **Waterflow Switches:** On all sprinkler system zones (wet pipe, preaction, dry pipe) - **Tamper Switches:** On all control valves, preaction valves, fire pump isolation valves - **Pressure Switches:** High/low air pressure on preaction and dry pipe systems - **Fire Pump Supervision:** Running, power loss, controller trouble

3.9.6 Notification Appliances

A. Audible Notification - **Type:** Speaker-strobes (voice-capable) or horn-strobes - **Coverage:** All occupied areas per NFPA 72 (15 dBA above ambient sound) - **Evacuation Signal:** Temporal 3 pattern (ANSI S3.41) or voice message

B. Visual Notification - **Type:** Xenon or LED strobes, wall-mounted - **Spacing:** Per NFPA 72 spacing tables (based on room size and candela rating) - **ADA Compliance:** Synchronized flashing, minimum 15 cd in public areas, 177 cd in high ambient light

C. Voice Evacuation / Mass Notification - **System Type:** Integrated emergency communication system (combination fire/mass notification) - **Capability:** Live paging via microphone at FACP or remote annunciators - **Pre-Recorded Messages:** Customizable emergency messages (fire evacuation, severe weather, security threat) - **Zone Control:** Individual notification zones for targeted messaging - **Notification Zones:** - MMR-A - MMR-B - IXP Equipment Room - Office and Support Areas - NOC/SOC - Loading Dock - Exterior Areas (if speakers provided)

3.9.7 System Wiring and Survivability

A. Signaling Line Circuits (SLC) - **Class:** Class A (redundant loop) for critical areas (MMRs, IXP) - **Class:** Class B (single path) for office and support areas - **Pathway:** Dedicated fire alarm conduit (no other system wiring)

B. Notification Appliance Circuits (NAC) - **Class:** Class B - **Supervision:** End-of-line resistors - **Pathway:** Dedicated conduit

C. Fiber Optic Backbone (Between FACP's) - Class: Class X per NFPA 72 (diverse routing, pathway separation >3 ft or fire-rated barrier) - **Media:** Single-mode fiber optic cable - **Survivability:** Two diverse fiber paths between main and sub-panels

D. Fire Alarm Power - Primary Power: Dedicated 120V circuit from house UPS panel (4-hour backup minimum) - **Secondary Power (FACP Battery):** 24-hour standby + 15-minute alarm (NFPA 72 minimum) - **Battery Type:** Sealed lead-acid or lithium iron phosphate

3.9.8 Sequence of Operations

A. Standard Fire Alarm Activation (Wet Pipe Areas) 1. Smoke detector or manual pull station activates 2. FACP receives alarm signal 3. Audible/visual notification appliances activate (temporal 3 pattern or voice message) 4. HVAC systems shutdown (via BMS interface) 5. Elevator recall (if applicable) 6. Magnetic door holders release 7. Notification to central monitoring station

B. Preaction Release Sequence (MMR-A, MMR-B, IXP Equipment Room) 1. **Stage 1 - Single ASD Detection (Alert/Action/Fire 1):** - Local alarm notification in affected zone - Alert to SOC and NOC - No water release, investigation required

2. **Stage 2 - Cross-Zone Detection (ASD Fire 2 + Spot Smoke Detector):**

- Preaction valve solenoid energizes
- Water fills piping up to closed sprinkler heads
- "Pre-Action System Activated" announcement
- Full alarm notification to building occupants
- Notification to fire department

3. **Stage 3 - Sprinkler Head Activation (Heat):**

- Sprinkler head fuses open
- Water discharges
- Waterflow switch activates
- Full evacuation signal

C. Supervisory and Trouble Signals - Supervisory (Yellow): Valve tamper, low air pressure (preaction), fire pump status - **Trouble (Yellow):** Ground fault, circuit open, power loss, device missing - **Alarm (Red):** Smoke, heat, manual pull station, waterflow

D. System Reset - After alarm investigation and clearance by fire department - Reset at main FACP (key-operated switch) - All notification appliances silence - System returns to normal supervisory mode

3.9.9 Testing and Maintenance

A. Acceptance Testing - Smoke detectors: Aerosol smoke or magnet testing - ASD system: Calibration and sensitivity verification - Manual pull stations: Operation verification - Notification appliances: Sound pressure level measurements (dBA) and strobe

synchronization - Preaction valve release: Simulated trip test (coordination with sprinkler contractor) - Voice evacuation: Intelligibility testing (STI measurements)

B. Ongoing Maintenance (per NFPA 72 and NFPA 25) - Quarterly: Notification appliance testing (visual inspection) - Semi-Annual: Detector testing (sample basis with rotating schedule) - Annual: Full system testing including preaction valve trip test - ASD filters: Cleaned per manufacturer recommendation (typically quarterly)

C. Inspection and Testing Records - Maintained onsite for AHJ review - Electronic records in BMS/SCADA system

3.9.10 Monitoring and Reporting

A. On-Site Monitoring - Main FACP in fire riser room (24/7 access for fire department) - Annunciator in Security Operations Center (24/7 staffed monitoring)

B. Off-Site Monitoring - Alarm, supervisory, and trouble signals transmitted to UL-listed central monitoring station - Communication path: Dual-path (IP network + cellular backup) - Automatic fire department dispatch upon alarm signal

C. Integration with Building Systems - BMS: System status, alarm notification, HVAC shutdown commands - EPMS: Alarm notification - Security System: Door release, lockdown coordination

3.10 SECURITY

3.10.1 Overview

The Carrier Hotel security system implements a Defense-in-Depth strategy with layered physical and electronic controls. The design recognizes the facility's dual requirements: (1) providing 24/7 carrier technician access for routine maintenance, and (2) maintaining the highest security standards for telecommunications infrastructure and tenant equipment.

Security Layers: - Layer 5: Public (perimeter fence, vehicle access control) - Layer 4: Building Exterior (access control at entries) - Layer 3: Building Interior (corridor access) - Layer 2: Meet-Me Rooms (MMR-A, MMR-B, IXP) - Enhanced Security - Layer 1: Individual Carrier Cages - Tenant-Controlled Access

3.10.2 Applicable Codes, Standards, and Guidelines

- NFPA 730: Guide for Premises Security (2023)
- NFPA 731: Standard for the Installation of Premises Security Systems (2023)
- ANSI/TIA-5017: Telecommunications Physical Network Security Standard (2016)
- ANSI/BICSI 002-2019: Data Center Design and Implementation Best Practices
- Open Compute Data Center Physical Security Guidelines Version 1.0
- CISA Critical Infrastructure Security Guidelines

3.10.3 General Security Systems

A. Access Control System (ACS)

System Type: - Enterprise-level, on-premises access control system - Supports 10,000+ cardholders, 500+ readers - Multi-tier access levels with time-based and zone-based restrictions - Anti-passback capability (prevent credential sharing) - Integration with video surveillance for visual verification - Biometric capability for high-security areas

Head-End Equipment: - Primary and backup servers in Security Operations Center (SOC) - SQL database with redundant backup - Network-based architecture (all devices on IP network) - Web-based administration interface

Readers and Credentials: - **Card Readers:** Contactless smart card (13.56 MHz, encrypted), mullion or gang-box mount - **Biometric Readers:** Fingerprint or iris recognition at MMR entries - **Credentials:** Smart cards with photo ID, proximity badges for lower-security areas - **Visitor Credentials:** Temporary cards issued at reception, time-limited access

Door Hardware: - **Electric Strikes or Mag Locks:** Fail-safe (unlock on power loss for egress, re-lock on restore) - **Request-to-Exit (REX) Devices:** Motion sensors or touchless switches on secure side - **Door Position Switches:** Magnetic contacts on all controlled doors (monitor open/closed status) - **Anti-Tailgate Detection:** Turnstiles or optical beam arrays at high-security entries

Access Zones: - **Public:** Perimeter and parking areas (no access control) - **Reception/Lobby:** Card reader + visitor management - **Carrier Tech Entry:** 24/7 card reader access for credentialed carrier technicians - **Office Areas:** Card reader access during business hours - **NOC/SOC:** Card reader + keypad (dual-factor authentication) - **Meet-Me Rooms (MMR-A, MMR-B, IXP):** Card reader + biometric reader (triple-factor with escort) - **Individual Carrier Cages:** Tenant-provided locks (3-point locking, tenant controlled)

Visitor Management: - Visitor check-in kiosk at main reception - Photo capture and badge printing - Pre-registration capability for expected visitors - Escort requirements enforced via access control programming (visitor badges do not open doors without escort present)

B. Video Surveillance System (VSS)

System Type: - IP-based megapixel camera system - Network Video Recorder (NVR) with redundant storage - 180-day retention minimum (90-day onsite + offsite backup) - Remote viewing capability via secure web portal - Video analytics: Motion detection, loitering alerts, object removal detection

Camera Coverage:

Location	Camera Type	Purpose
Building Perimeter (all sides)	Fixed bullet, 4 MP	Perimeter monitoring, trespassing detection
Parking Areas	PTZ, 4 MP	Vehicle and pedestrian monitoring, license plate capture

Location	Camera Type	Purpose
Main Entry	Fixed dome, 4 MP	Entry/exit monitoring, facial recognition-capable
Carrier Tech Entry	Fixed dome, 4 MP + LPR camera	24/7 technician entry/exit, vehicle tracking
Loading Dock	Fixed bullet, 4 MP	Delivery monitoring, dock activity
Lobby and Reception	Fixed dome, 4 MP	Visitor management, general surveillance
Corridors (all)	Fixed dome, 4 MP	Interior circulation, incident response
MMR-A Entry	Fixed dome, 4 MP	Access verification, multi-angle view
MMR-B Entry	Fixed dome, 4 MP	Access verification, multi-angle view
MMR-A Interior	PTZ, 4 MP (3 cameras)	Equipment monitoring, incident investigation
MMR-B Interior	PTZ, 4 MP (3 cameras)	Equipment monitoring, incident investigation
Cross-Connect Fields	Fixed dome, 4 MP	Fiber work monitoring, troubleshooting
IXP Equipment Room	Fixed dome, 4 MP	Equipment protection
NOC	Fixed dome, 4 MP	Operator workstation monitoring
SOC	Fixed dome, 4 MP	Security staff monitoring, audit trail
Electrical Rooms	Fixed bullet, 2 MP	Infrastructure protection

Camera Specifications: - **Resolution:** Minimum 4 MP (2688 x 1520) for critical areas, 2 MP for secondary areas - **Frame Rate:** 30 FPS (critical areas), 15 FPS (secondary areas) - **Low Light:** 0.01 lux color performance, automatic IR illumination - **Compression:** H.265 (HEVC) for bandwidth efficiency - **PoE:** 802.3at (PoE+) power via network switch - **Analytics:** Onboard video analytics (motion detection, object classification)

Recording and Storage: - **NVRs:** Redundant servers in SOC (N+1 configuration) - **Storage:** RAID 6 hard drive array, 180-day retention at design resolution/frame rate - **Offsite Backup:** Automatic cloud backup of critical camera streams - **Playback:** Multi-camera simultaneous playback, event-based search, export capability

C. Perimeter Intrusion Detection System (PIDS)

System Type: - Fence-mounted fiber optic or buried cable detection - Integration with VSS for automatic PTZ camera tracking - GPS coordinate reporting for precise intrusion location - Weatherproof, low false-alarm rate

Coverage: - Entire perimeter fence line - Zoned detection (50-meter zones typical) - Automatic alert to SOC with map location - PTZ camera slew to intrusion location

Detection Technology: - **Fence-Mounted:** Fiber optic cable detecting vibration and cutting attempts - **Buried Cable:** RF field disturbance detection in buried sensor cable - **Analytics:** Machine learning to discriminate between human, animal, and environmental triggers

D. Intrusion Detection (Building Interior)

Coverage: - After-hours intrusion detection in unmanned areas - Motion detectors in storage areas, electrical rooms, mechanical rooms - Glass-break detectors at exterior windows - Door contacts on all exterior doors (monitored 24/7)

Operation: - Armed during unoccupied hours (programmable schedule) - Integrated with access control system (automatic disarm on authorized entry)

3.10.4 Security Operations Center (SOC)

A. Physical Layout - Dedicated room adjacent to Network Operations Center - Secure entry: Card reader + keypad - Area: 800 SF - Operator positions: 4 workstations (24/7 staffing)

B. Equipment - Video wall: 3x3 LCD display array (monitors all camera feeds) - Operator workstations: Dual monitors per position - Access control head-end servers (primary + backup) - Video management servers (primary + backup) - Fire alarm annunciator - Intrusion detection panel - PIDS monitoring station - UPS backup power (4 hours)

C. Functions - 24/7 monitoring of all security systems - Alarm response and investigation - Visitor management and escort coordination - Incident logging and reporting - Integration with fire alarm and building management systems

3.10.5 Security Integration with Building Systems

A. Fire Alarm Integration - Fire alarm status displayed on SOC video wall - Automatic door unlock on fire alarm (all controlled doors fail-safe) - Video surveillance continues to record during emergency - Access control log preserves records of personnel locations during evacuation

B. Building Management System (BMS) Integration - Security system status reported to BMS (normal/alarm/trouble) - HVAC control during security incidents (smoke purge, lockdown mode)

C. Elevator Control (If Applicable) - Card reader in elevator cab for floor access control - Elevator recall on fire alarm or security lockdown

3.10.6 Cybersecurity (Physical Systems)

A. Network Segmentation - Security system on dedicated VLAN (isolated from IT network) - Firewall protection at network boundary - No direct internet access (all remote access via VPN)

B. Access Control - Role-based access to security systems (administrator, operator, view-only) - Multi-factor authentication for system administrators - Audit trail of all system access and configuration changes

C. Encryption - AES-256 encryption for credential data - TLS encryption for all network traffic (cameras, access control, video management)

3.10.7 Physical Security - Meet-Me Room (MMR)

The MMR is the facility's highest security zone, requiring enhanced measures:

A. Construction - Slab-to-slab partitions (prevent over-the-top intrusion) - Walls: CMU or steel stud + 5/8" Type X GWB each side - Doors: Hollow metal, 3' x 7', 3-point locking - Vision panels: Ballistic-rated glazing (UL 752 Level 1 minimum)

B. Access Control - Entry Sequence: 1. Card reader (verify credential + time/day authorization) 2. Biometric reader (fingerprint or iris scan) 3. PIN keypad (knowledge factor) 4. Escort verification (SOC confirms escort present if required) - **Anti-Tailgate:** Turnstile or mantrap vestibule preventing multiple entries on single credential - **Egress:** REX device on secure side allows free egress (life safety)

C. Surveillance - Multi-angle camera coverage (no blind spots) - Camera views of all entry/exit, aisles, cross-connect fields - 4 MP resolution, 30 FPS frame rate - 180-day retention

D. Intrusion Detection - Motion detectors (armed after-hours when MMR unoccupied) - Door position monitoring (alarm if door open without valid access event)

E. Carrier Cage Security - Individual cages with steel mesh (4-gauge, <0.5" openings) - Tenant-provided locks (tenant controls access to own cage) - Carrier responsibility for cage key management - Optional: Card reader integration for carriers desiring electronic access control

3.11 TELECOMMUNICATIONS

3.11.1 Overview

The Carrier Hotel telecommunications infrastructure represents the facility's core value proposition, designed as a carrier-neutral environment supporting diverse fiber entry points, high-density cross-connect capability, and optimal routing for minimizing latency and maximizing redundancy. The design accommodates 11,600 fiber strands from 13+ major carriers while providing flexible pathways for future carrier additions.

3.11.2 Codes and Standards

- ANSI/TIA-942: Telecommunications Infrastructure Standard for Data Centers (current edition)
- ANSI/TIA-568: Generic Telecommunications Cabling for Customer Premises
- ANSI/TIA-569: Commercial Building Standard for Telecommunications Pathways and Spaces

- ANSI/TIA-606-C: Administration Standard for Telecommunications Infrastructure
- ANSI/TIA-607-B: Generic Telecommunications Bonding and Grounding for Customer Premises
- ANSI/TIA-758: Customer-Owned Outside Plant Telecommunications Infrastructure Standard
- BICSI TDMM: Telecommunications Distribution Methods Manual (15th Edition)
- BICSI 002: Data Center Design and Implementation Best Practices

3.11.3 Service Entrances and Carrier Diversity

A. Diverse Fiber Entry Points - Minimum: Two geographically diverse points of entry (POE) - **Separation:** >100 feet separation between POE locations - **Entry Method:** Underground conduit to below-grade vaults, then interior fiber entry rooms - **Redundancy Goal:** No single fiber cut can disable all carrier connectivity

B. Fiber Entry Rooms (FER) - Quantity: (2) Fiber Entry Rooms (FER-A, FER-B) - **Location:** Opposite ends of building exterior wall - **Size:** 500 SF each - **Function:** Transition from outside plant (OSP) to inside plant (ISP) fiber - **Equipment:** - Fiber splice enclosures - Slack storage - Cable management hardware - Fiber distribution panels

C. Carrier Duct Bank Infrastructure - Configuration: Multiple 4" or 6" conduits per carrier in concrete-encased duct bank - **Routing:** From property line to building FER via diverse paths - **Capacity:** Designed for 13 carriers initially, expandable to 20+ - **Vault System:** Carrier-accessible vaults at property line for splice access - **Separation:** Minimum 10' horizontal separation between diverse duct banks

D. Carrier Neutral Model - Open access: No exclusive arrangements with any carrier - Equal treatment: All carriers offered equivalent terms, pathways, and space - Tenant choice: Tenants select carriers independently

3.11.4 Pathways and Cable Management

A. Backbone Pathways (FER to MMR)

Fiber Pathway Design: - **Primary Route:** Dedicated cable tray from FER-A to MMR-A, FER-B to MMR-B - **Cross-Connect Route:** Tray between MMR-A and MMR-B for carrier interconnection - **Tray Type:** Ladder-type cable tray, 24" wide minimum - **Capacity:** Design for 50% fill at initial installation (100% fill at future buildout) - **Pathway Separation:** Physically diverse routes for redundant fiber paths (maintain >3' separation)

Fiber Management: - Cable tray with waterfall adapters at vertical transitions - D-ring cable management for organized routing - Minimum bend radius: 15x cable diameter (1.5" typical for multi-fiber cables) - Cable ties: Velcro or releasable for future adds/moves/changes - Identification: Labeling per TIA-606-C at 10' intervals and at all access points

B. Horizontal Pathways (Within MMRs)

Cross-Connect Field Support: - Overhead cable tray above fiber termination racks - Vertical cable managers at each rack - High-density fiber patch panels (up to 144 fibers per panel) - Front-access design for efficient patching

Carrier Cage Pathways: - Cable tray extending into each carrier cage from main distribution tray - Segregated pathways: Customer fiber separate from facility fiber - Adequate capacity: Design for 3x initial fiber count per cage

C. Copper Pathways

Management Network: - Separate cable tray for Cat 6A copper infrastructure - Out-of-band management network for carrier equipment - Minimum 12" separation from power cables (18" preferred)

Pathway System: - Dedicated telecom cable tray (no power cables in same tray) - Ladder tray above raised floor in MMRs - J-hooks or cable tray in office areas above ceiling

3.11.5 Meet-Me Room (MMR) Infrastructure

A. MMR-A and MMR-B Layout

Space Allocation: - Total area per MMR: 12,000 SF - Carrier cage area: 8,000 SF (approximately 80 cages at 100 SF each) - Cross-connect field area: 3,000 SF - Circulation and equipment staging: 1,000 SF

Carrier Cages: - Modular cage sizes: 100 SF, 200 SF, 400 SF, 800 SF - Welded steel mesh construction (4-gauge, <0.5" openings) - Full-height (floor to deck above) - Door: Steel mesh, 3-point locking, customer-controlled access - Power: Dual-feed circuits (A-side and B-side) to customer panel - Cooling: Shared precision cooling infrastructure (design for 8 kW/rack average per cage)

Cross-Connect Fields: - High-density fiber termination racks - Layout: Hot aisle / cold aisle configuration (airflow management even though minimal heat load) - Fiber panels: 96-strand termination panels (LC or MTP connectors) - Density: 48 panels per relay rack, 4,608 fiber termination points per rack - Quantity: 20+ racks per MMR supporting >90,000 termination points - Labeling: Per TIA-606-C (rack, panel, port-level identification)

B. Fiber Distribution Frames (FDF)

Main Distribution Area (MDA): - Centralized fiber distribution frames in each MMR - Aggregation point for all carrier fibers entering the MMR - Backbone to Intermediate Distribution Areas (IDAs) within the same MMR

Intermediate Distribution Area (IDA): - Distributed fiber panels throughout MMR serving specific zones - Connects carrier cages to cross-connect fields - Supports efficient horizontal fiber routing

C. Fiber Types and Standards

Single-Mode Fiber (SMF): - **Type:** OS2 single-mode fiber (9/125 µm) - **Application:** All inter-building backbone, long-distance carrier connections - **Connector:** LC duplex (primary), MPO/MTP for high-density applications - **Standards:** TIA-568.3-D

Multimode Fiber (MMF): (If required for short intra-building links) - **Type:** OM4 multimode fiber (50/125 μm) - **Application:** Short links within building (<300 meters) - **Connector:** LC duplex - **Standards:** TIA-568.3-D

D. Copper Infrastructure (Management Network)

Category 6A Cabling: - **Application:** Out-of-band management, console access, building services - **Standard:** TIA-568.2-D - **Connector:** RJ45, modular jacks - **Testing:** Certified to Category 6A performance (500 MHz, 10GBASE-T)

Telecommunication Rooms: - Main Distribution Frame (MDF): Located in FER or MMR - Intermediate Distribution Frames (IDFs): Located in office areas, NOC, SOC - **Equipment:** Network switches, patch panels, cable management

3.11.6 Internet Exchange Point (IXP) Infrastructure

A. IXP Equipment Room

Purpose: - Dedicated space for IXP switching fabric - Neutral exchange point allowing multilateral peering - Operated by IXP operator (e.g., DE-CIX partnership) or Nexus directly

Equipment: - IXP switching fabric: High-capacity Ethernet switches (100G, 400G capable) - Route servers: For BGP route exchange between participants - Looking glass servers: Public visibility into IXP routing - Monitoring systems: Traffic statistics, performance monitoring

Connectivity: - Fiber connections from IXP switches to each participating carrier's equipment in MMRs - Redundant paths: Each participant connects via diverse fiber routes (A-side and B-side)

B. IXP Services

Public Peering: - Participants connect to shared switching fabric - Multilateral peering via route servers - Single connection provides peering with all IXP participants

Private Interconnection: - Direct cross-connects between carriers (bypassing IXP switch) - Fiber patch between carrier cages in MMR - Managed through cross-connect field or direct cage-to-cage fiber

3.11.7 Telecommunications Grounding and Bonding

A. Telecommunications Grounding Infrastructure (per TIA-607-B)

Telecommunications Grounding Busbar (TGB): - Location: MMR-A, MMR-B, FER-A, FER-B, IDF rooms - Specifications: 1/4" x 4" copper bus, tin-plated, minimum 10 feet long - Mounting: Insulated standoffs, accessible for bonding connections - Bonding to Building Ground: #3/0 AWG bare copper conductor to nearest electrode

Telecommunications Bonding Backbone (TBB): - **Purpose:** Interconnect TGBs throughout facility - **Conductor:** #3/0 AWG bare copper minimum - **Routing:** Dedicated pathway separate from power distribution - **Connections:** Exothermic weld or irreversible compression

Equipment Bonding: - **Rack Grounding:** Each equipment rack bonded to TGB via #6 AWG insulated copper conductor - **Cable Tray Bonding:** Cable tray system bonded to building ground at 50' intervals - **Overhead Ground Bus:** #2 AWG bare copper bus in cable tray, bonded to TGB

B. Lightning Protection Coordination

Fiber Entry Protection: - Non-metallic fiber cables (no metallic strength members) - Grounding of cable support structures (J-hooks, cable tray) - Bonding of fiber entry room equipment to TGB

Copper Cable Protection: (if any copper enters from outside) - Primary protection: Gas tube surge protectors at entry point - Secondary protection: Solid-state surge protectors at equipment - Grounding: All protectors bonded to TGB

3.11.8 Cable Management and Labeling

A. Cable Identification (per TIA-606-C)

Label Format: - Building-Floor-Room-Rack-Panel-Port - Example: "CH-1-MMRA-R05-P12-08" (Carrier Hotel, Floor 1, MMR-A, Rack 5, Panel 12, Port 8)

Label Locations: - Both ends of each cable - At all splice points and patch panels - On cable tray at access points

Label Materials: - Permanent adhesive labels (laser-printable) - Wrap-around labels for cables - Tamper-evident labels for secure cross-connects

B. Cable Records

Documentation: - As-built cable routing drawings - Fiber count and type per cable - Splice records showing fiber assignments - Cross-connect database (electronic records)

Database System: - Centralized cable management system - Integration with access control (track who accesses which cages and racks) - Change management: All cross-connects logged with date, technician, authorization

3.11.9 Testing and Acceptance

A. Fiber Optic Testing

Installation Testing: - **OTDR Testing:** Optical Time Domain Reflectometer test on all backbone fibers - **Insertion Loss:** Measured end-to-end, compared to calculated loss budget - **Acceptance Criteria:** ≤ 0.35 dB per connector, ≤ 0.5 dB per splice (fusion splice)

Certification: - Tier 1 Certification (basic): OTDR trace, insertion loss - Tier 2 Certification (extended): Includes return loss, polarization mode dispersion

B. Copper Testing

Category 6A Testing: - Fluke DTX or equivalent tester - Parameters: Wire map, length, insertion loss, return loss, NEXT, FEXT, alien crosstalk - Certification: Pass TIA-568.2-D Category 6A specifications

C. Pathway Testing

Cable Tray Inspection: - Verify separation from power cables - Verify adequate support spacing and grounding - Confirm fire-stop penetrations properly sealed

DOCUMENT COMPLETION

This Carrier Hotel Basis of Design provides comprehensive technical criteria for all building systems necessary to deliver a world-class telecommunications facility serving as the Nexus Hubbard Campus network gateway, private peering hub, and Internet Exchange Point.

Key Design Outcomes: - 487,500 SF facility supporting 3,500+ network equipment cabinets - 50 MW IT load with N+1 electrical redundancy - 11,600 fiber strands from 13+ carriers with geographic diversity - Meet-Me Room infrastructure supporting 500+ tenant organizations - Phased construction deployment (5 phases × 10 MW IT load per phase) - Precision environmental control (72°F ±2°F, 40-50% RH) - Comprehensive security with Defense-in-Depth architecture - Carrier-neutral open access model - Integration with campus SCADA and building management systems

This document serves as the foundation for detailed design development and construction documentation. All subsequent design work shall conform to the criteria established herein unless formally modified through the design change process with owner approval.